Udyam Digital — Phase 1 Business Intelligence, Product Strategy & System Design

YTL Hackathon 2026 (Airtel SYNC) · Enterprise Track A · Project Udyam Digital

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How to read this document

This PDF has two parts. **Part A (Sections 1–20)** is the Phase 1 business intelligence and product strategy report. **Part B (D1–D7)** is the system design (`docs/DESIGN.md`) built on its findings.

This report turns the supplied data into a product decision. It moves in one direction: **raw data → analysis → findings → business implications → opportunities → recommended solution → build priorities.**

Every number has a source tag. The tag points to the file in the repository that produced it, so anyone can re-run and check it.

Tag	Source file	What it contains
[P]	`analysis/01_profile.md`	Data quality, funnel check, registry joins
[S]	`analysis/02_signals.md`	Conversion, drop reasons, churn, bundles, lead sources (Q1–Q7)
[B]	`analysis/bundles_and_metrics.md`	Segment ranking, validated starter bundles, metric baselines and targets
[M]	`analysis/03_propensity.md`	Lead-scoring model results
[F]	`analysis/05_report_facts.md`	Segment profiles, source quality, personas, KYC-stage baselines
[K]	`analysis/kyc/kyc_results.md`	OCR and verification results for applicants K01–K12
[Case]	`Airtel_SYNC_Case.pdf` and dataset README	Figures stated by the organisers

Four kinds of statement are kept apart:

- **Evidence**: what the data shows, with a number and a tag.
- **Interpretation**: what we think the evidence means.
- **Assumption**: something we believe but cannot prove from this data.
- **Recommendation**: what we propose to build or do.

Where it matters, a statement is labelled with one of these words in bold.

Plain-English glossary

Term	Meaning
Lead	A business that showed interest (filled a form, was visited, was referred)
Activation	The business is live and paying
Conversion	Share of leads that become activations
CAC	Customer acquisition cost: sales and partner money spent per activated customer
Churn	Customer leaves within 6 months
NPS	Satisfaction score, 0–10, given by activated customers
KYC	“Know your customer”: GST certificate, Udyam certificate, signatory ID and address proof
GSTIN	15-character GST registration number; it has a built-in check digit
Udyam	Government MSME registration number
Propensity score	A model’s estimate of how likely a lead is to activate
AUC	How well a model ranks good leads above bad ones (0.5 = coin toss, 1.0 = perfect)
Lift	How much more often two products are bought together than chance would predict

Section 1 — Executive Summary

What we are solving. Accel Business (a fictional stand-in for Airtel Business) sells connectivity, communication, cloud, security and IoT products to small businesses: shops, clinics, CA firms, restaurant chains and small factories. Today it sells them one product at a time through a field sales force built for large companies. The case asks for a digital journey that lets an owner discover the right bundle, buy it and get it activated **on a phone, in their own language, with minimal human effort** [Case].

Who the customers are. Across 5,000 leads from one quarter, the split is:

Segment	Share of leads	Typical example
Micro retail & services	54.2%	kirana, salon, bakery
Professional offices	24.9%	law office, coaching centre, diagnostic lab
Small multi-site businesses	15.1%	hotel group, retail chain
Small manufacturers	5.8%	moulding, packaging, textile units

Source: [F]. This closely matches the case's indicative split of 55/25/15/5 [Case].

What the journey looks like today. A lead is contacted, a salesperson visits to assess needs, a quote is prepared, KYC documents are collected, and the service is installed. For customers who made it through, the steps took a median of 2, 3, 4, 4 and 4 days. Activation took **17.6 days** on average, and customers chased status **3.0 times** [S].

Where it breaks. Only **368 of 5,000 leads (7.4%)** became customers [P]. The weakest step is the quote: only **43.6%** of assessed leads accept it [F], and for micro businesses it's 32% [S]. The single largest group of losses is about reach and speed rather than price. Unreachable leads, owners with no time for a visit, visits never scheduled, slow quotes and late installs together cause **41.2%** of all drops [S].

What the data says the opportunity is.

- **Evidence.** About **32%** of today's drops have causes that a self-serve digital journey removes outright, and another **37%** have causes it partly addresses [S].
- **Evidence.** Sellers today sold **56 of 368** customers (15%) products the catalogue rules forbid, such as GST-only products on cancelled or missing GSTINs [S].
- **Evidence.** Customers who chased status five or more times churned at **28%**, against **6%** for those who chased once [S].
- **Evidence.** Referrals convert at **14.8%** for **Rs 4,099** per activation, while field sales converts at 7.3% for **Rs 24,335** [S].

What the team should focus on. We recommend a hybrid product (Section 13). An AI advisor and a rules engine handle discovery, pricing and KYC for the common case. A live tracker removes the need to chase. Exceptions go to a human with full context: KYC flags, delays, and multi-site deals. Launch with **professional offices** and treat multi-site businesses as an assisted track (Section 5).

Five takeaways every teammate must understand

1. **The quote step is the bottleneck.** Only 43.6% of assessed leads accept a quote, and just 32% for micro [F][S]. A clear, instant, plain-language price is the single most valuable screen we build.
2. **Speed beats price.** 41.2% of drops are about reach and speed, 23.9% about understanding and price, and 5.9% about paperwork [S]. Remove the wait and you remove the biggest leak.
3. **The advisor must enforce the catalogue rules.** Today, 15% of sold bundles break them [S]. The judges will test eligibility live, with 15 hidden scenarios [Case]. Rules decide eligibility; the AI only explains.
4. **The tracker is a retention feature.** Churn climbs from 6% to 28% as chase calls rise from 1 to 5+, and NPS falls from 7.5 to 6.0 [S].
5. **KYC is mostly lookups, but it needs a human safety net.** 6 of the 12 sample applicants have planted problems that the checker caught. Two of them can't be resolved by software alone: a GSTIN digit that differs from the registry, and a legal name that names a different business [K].

Section 2 — Understanding the Business Problem

2.1 The current journey, stage by stage

Stage	What happens	What the customer experiences	What the business needs	What can go wrong	What the data shows
Lead → Contacted	Sales or a partner tries to reach the lead	Waits for a call that may come at a bad time	Reach the owner while interest is fresh	Unreachable, wrong number, not interested	1,594 leads lost (34.4% of all losses); 68.1% reached [F]. Unreachable after 3 attempts is the top drop reason overall (17.4%) [S]
Contacted → Needs assessed	A salesperson schedules a visit to understand needs	Has to find time for a meeting	Understand the business well enough to propose a bundle	Owner has no time; visit never scheduled; locked in with another provider	1,432 lost (30.9%); 58.0% pass. “No time for visit” 531 and “visit not scheduled” 296 [S]
Needs assessed → Quote accepted	A quote is prepared and sent	Waits 3–5 days [Case] for a PDF with product codes	A price the owner understands and accepts	Too expensive, too complex, too slow, competitor cheaper	1,113 lost (24.0%); **43.6% pass, the weakest step** [F]
Quote → KYC complete	GST, Udyam, signatory ID and address proof are collected	Hunts for documents and sends them physically or by email	Verified identity, compliant with the DPDP Act	Missing documents, paperwork fatigue, signatory unavailable, mismatched details	273 lost; all 273 are paperwork reasons [F]. 68.3% pass
KYC → Activated	Installation and first payment	Waits for installation and chases status	Install fast, collect the first payment	Install late (over 21 days), cancellation, site not feasible for fibre, failed payment	220 lost: 98 late installs, 53 cancellations, 41 not feasible, 28 payment failures [F]

2.2 Symptoms and root causes

The data records **symptoms** (the drop reasons). We have to infer the **root causes**.

Symptom (recorded)	Leads	Likely root cause (interpretation)
Unreachable after 3 attempts; wrong number	805 + 319	The journey waits for the business to call the customer. Digital leads sit idle: 44% of WhatsApp-campaign drops and 34% of website-form drops are unreachable or wrong numbers [S]
Owner had no time for a visit; visit not scheduled	531 + 296	Discovery needs a physical meeting with a salesperson
Quote took too long; quote too complex	179 + 206	Quotes are made by hand and use product codes
Price too high; competitor cheaper; only wanted a mobile plan	414 + 207 + 280	Bundles are proposed without visible add-on choices, and some owners only wanted one product
Missing GST/Udyam; paperwork too long; signatory unavailable; details didn’t match	94 + 79 + 53 + 47	Documents are collected manually with no lookup or pre-fill, and checks happen late
Installation delayed past 21 days; cancelled before install	98 + 53	Physical install capacity, and no visibility for the customer while waiting

Counts are from [S].

2.3 Problems by type

Problem type	Where it shows up in the data	Size
Human dependency	Visit needed to assess needs; quote prepared by a person	No time for visit 531, visit not scheduled 296 [S]
Slow response	Unreachable leads; slow quotes; 9.0 days pass before a quote is accepted	805 unreachable; 179 slow quotes; 9.0 of the 17.6 days are pre-quote [S]
Complex pricing	Quote too complex; price too high	206 + 414 [S]
Documentation	All KYC-stage drops	273 [F]
Lack of customer understanding	Only wanted a mobile plan; 15% of bundles sold break the rules	280 drops; 56 mis-sold customers [S]
Poor follow-up	3.0 chase calls per customer; 68% chased 3 or more times	[F]
Installation delays	Installation delayed beyond 21 days	98 drops; 16% of activations took over 21 days [F]

Section 3 — Dataset Understanding and Quality

3.1 The five data sources

Dataset	Records	Purpose	Key columns	Connects to	Quality	Use in the product
`product_catalogue.csv`	25	What can be sold, at what price, to whom	price min/max, setup fee, minimum contract, city tiers, min employees, min sites, requires GSTIN, requires one of, install days, target segments	`bundle_products` and `products_of_interest` in leads	No duplicates; no min > max prices; every dependency points to a real product [P]	Advisor rules engine, pricing, tracker dates
`smb_leads.csv`	5,000	One quarter of leads with outcome	segment, source, size, language, GSTIN/Udyam, current internet and spend, stage dates, drop reason, bundle, value, churn, NPS, acquisition cost	GSTIN registry (`gstin`), Udyam registry (`udyam_number`), catalogue	40 consistency checks all returned 0; 124 blank `current_internet` (treated as Unknown) [P]	Analysis, propensity model, admin dashboard seed
`mock_gstin_registry.csv`	3,534	Backing data for a mock GSTIN lookup	gstin, legal name, trade name, constitution, address, state, registration date, status	Leads (every lead GSTIN is found)	3,266 Active, 167 Cancelled, 101 Suspended [P]	`GET /kyc/gstin/{gstin}` mock API
`mock_udyam_registry.csv`	2,723	Backing data for a mock Udyam lookup	udyam number, enterprise name, type, activity, state, city	Leads (every lead Udyam number is found)	Names, cities and states all match the leads [P]	`GET /kyc/udyam/{number}` mock API
`kyc_documents/`	47 images, 12 applicants	Specimen documents for OCR and validation	GST certificate, Udyam certificate, signatory ID, electricity bill	Registries, by GSTIN, Udyam number and name	One applicant (K05) has no Udyam certificate; several have planted problems [K]	Smart KYC test set

3.2 Data profiling results

Evidence [P]:

- **No duplicate rows** in any file, and no duplicate keys (`lead_id`, `product_id`, `gstin`, `udyam_number`).
- **Nulls only where expected.** `gstin` is blank for 1,466 leads with no GSTIN. `udyam_number` is blank for 2,277 leads with no Udyam. Stage dates are blank after the stage a lead reached. Activation fields are blank for the 4,632 dropped leads. `current_internet` and `current_provider` are blank for 124 leads.
- **Format checks all pass.** All GSTINs are 15 characters. All Udyam numbers match `UDYAM-XX-00-NNNNNNN`. City tier is always 1 or 2. NPS is always 0–10. No negative amounts.

- **Stage logic is consistent.** No stage date comes before the previous one. Every activated lead has an activation date and a bundle. Every dropped lead has a drop reason. `days_lead_to_activation` equals activation date minus creation date in every row.
- **Ranges:** created between 2026-04-01 and 2026-06-30; 1–150 employees; 1–12 sites; telecom spend Rs 200–27,750 a month.

3.3 Registry matching

Join	Result [P]
Leads with a GSTIN → GSTIN registry	3,534 of 3,534 found. 3,266 Active, 167 Cancelled, 101 Suspended
Leads with a Udyam number → Udyam registry	2,723 of 2,723 found. Name, city and state differ in 0 cases
Business name vs GST **legal** name	Differ in 2,443 cases, and every one is a proprietorship stored as "Owner (Prop. Trade name)"
Business name vs GST **trade** name	Differ in 0 cases
Lead state vs GST registry state	Differ in 0 cases. Each GSTIN state code maps to exactly one state
Activated leads with a Cancelled or Suspended GSTIN	22
Leads with Udyam but no GSTIN	636

Implication (recommendation): KYC name matching must compare against the registry's **trade name**. Matching on legal name would wrongly flag about 2,443 honest proprietors. Name-only lookups are also unsafe, because names repeat across cities (Section 9).

3.4 How the datasets relate

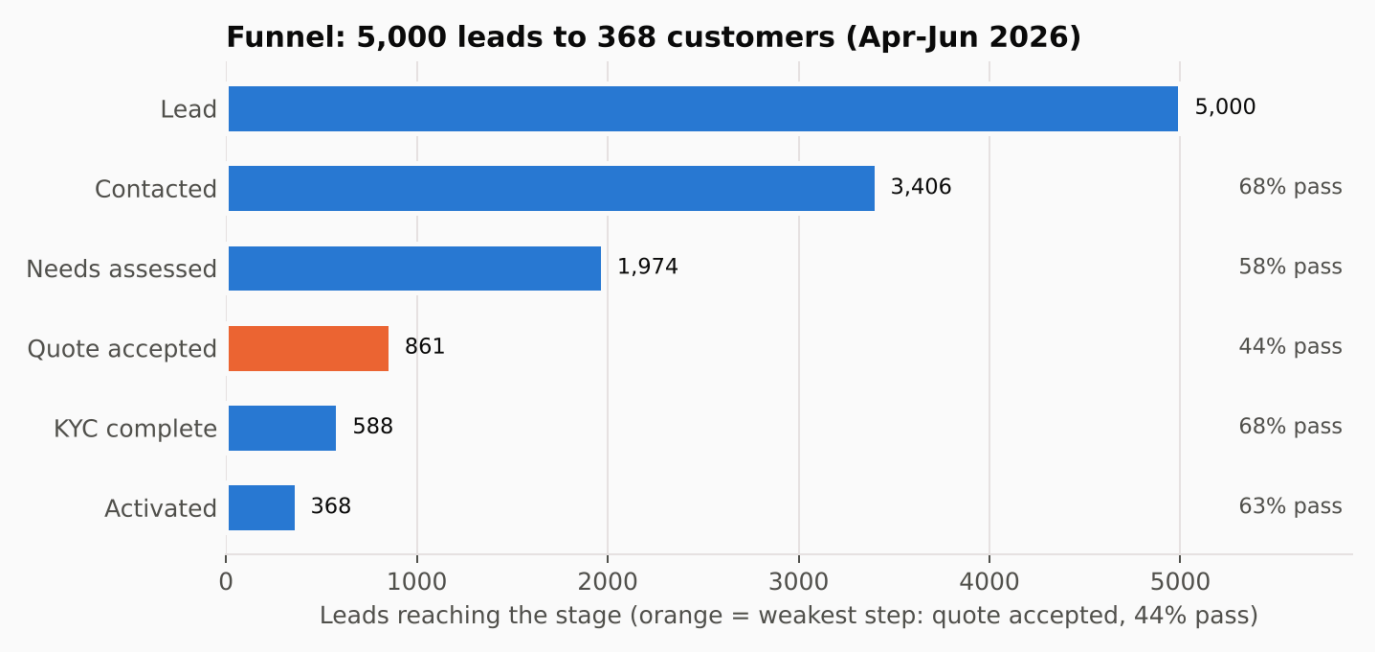
```
product_catalogue ──(product_id)── smb_leads.products_of_interest / bundle_products
smb_leads ──(gstin)──────────────── mock_gstin_registry
smb_leads ──(udyam_number)────────── mock_udyam_registry
kyc_documents ──(GSTIN, Udyam no., names read by OCR)── both registries
```

The KYC images aren't keyed to lead IDs. They link to the registries only through the numbers and names printed on them. This is the same situation the real product will face.

Section 4 — Complete Funnel Analysis

Stage	Leads	Conversion from previous stage	Drop-off at this step	Share of all losses
Lead	5,000	—	—	—
Contacted	3,406	68.1%	1,594	34.4%
Needs assessed	1,974	58.0%	1,432	30.9%
Quote accepted	861	**43.6%**	1,113	24.0%
KYC complete	588	68.3%	273	5.9%
Activated	368	62.6%	220	4.7%

Source: [F]. All six counts match the README exactly [P].



Interpretation: bars shrink fastest early on. In absolute terms the first contact loses the most people, but the quote step has the lowest pass rate (orange). Two-thirds of all losses (65.3%) happen before anyone has seen a price.

1. Where is the biggest loss?

- **Evidence.** In absolute numbers, at first contact: 1,594 leads, or 34.4% of all losses.
- **Evidence.** In pass-rate terms, at the quote: only 43.6% of assessed leads accept it [F].

2. Which stage has the most friction?

- **Interpretation.** The quote. It's where an owner first meets the price and the product codes, after waiting days for it. In micro retail only 32% pass this step [S].

3. What the numbers mean commercially.

- **Evidence.** Rs 5.54 million was spent acquiring these leads (sum of `acquisition_cost_inr`), which works out to Rs 15,053 per activation [S].
- **Interpretation.** Every lead lost before the quote carries its acquisition cost without ever reaching a price.

4. Which stages can automation improve?

- **Recommendation.** Contact, needs assessment and quote can be collapsed into one self-serve session, the advisor. These steps account for 89.3% of all losses.
- **Recommendation.** KYC (5.9% of losses, all paperwork) can be largely automated with registry lookups and OCR.

5. Which stages still need people?

- **Interpretation.** Installation, and the exceptions around it: 98 late installs, 41 sites not feasible for fibre, 53 cancellations [F].
- **Interpretation.** KYC cases that software can’t settle (Section 9).
- **Interpretation.** Multi-site deals with multi-week installs (Section 5).

Comparison with the case baseline. Our figures match the case: 7.4% conversion (the case says 7%), 17.6 days, 3 chase calls, CAC of about Rs 15,000, a Rs 6,900 median bundle, about Rs 2,500 for micro, and 9.6% vs 2.0% by GSTIN [Case][S].

One nuance is worth stating. The case calls quote acceptance the “steepest drop”. That’s true for the **pass rate** (43.6%). The largest **absolute** loss, though, is at first contact.

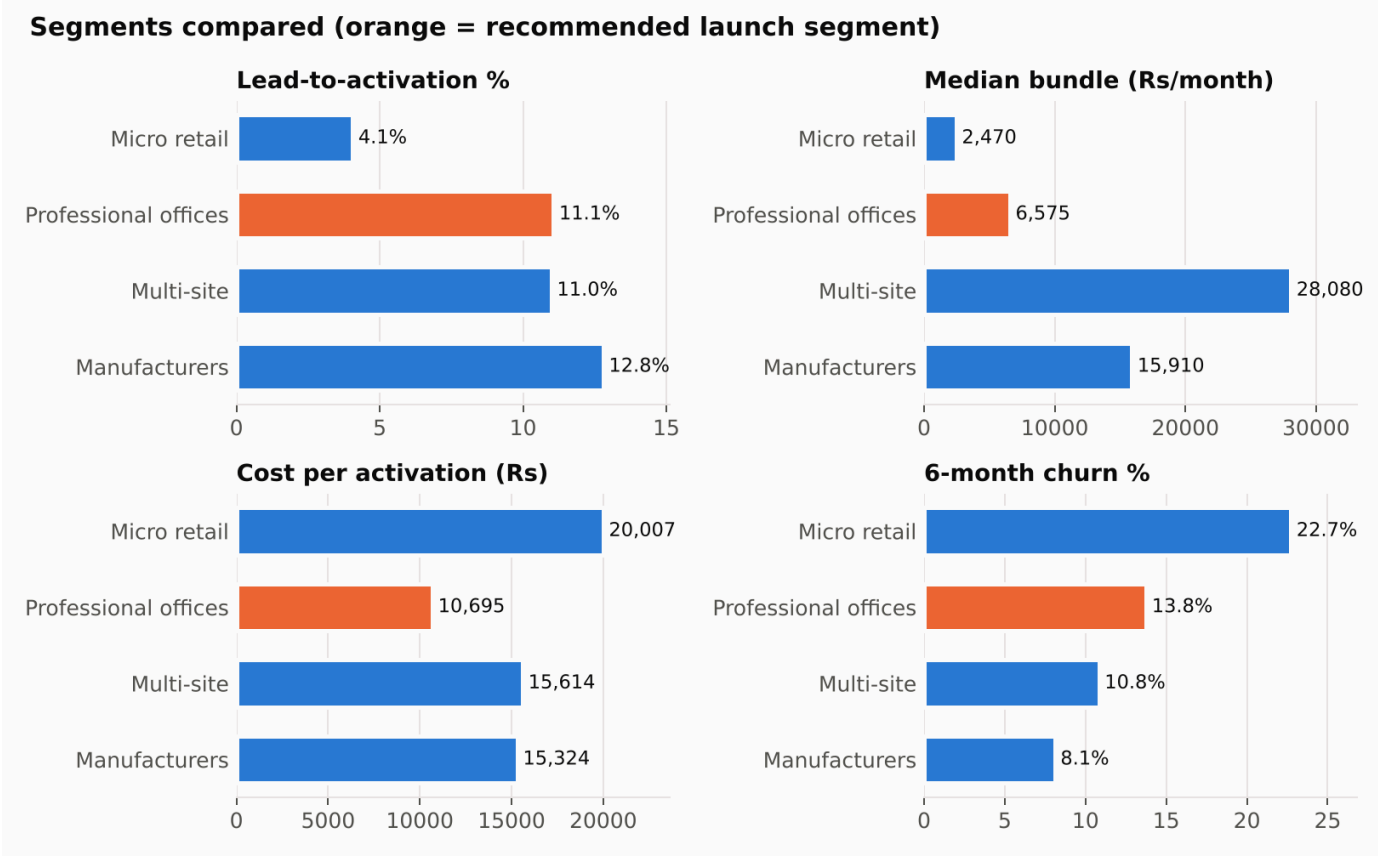
The case also describes installation and activation as taking 10–14 days. In the data, the step from KYC complete to activation takes a mean of 4.8 days. The 17.6 days is the whole journey from lead creation [S]. Both readings are consistent; they just measure different spans.

Section 5 — Customer Segment Analysis

5.1 Comparison table

	Micro retail & services	Professional offices	Small multi-site	Small manufacturers
Leads	2,709	1,247	755	289
Share of leads	54.2%	24.9%	15.1%	5.8%
Lead-to-activation	4.1%	11.1%	11.0%	**12.8%**
Most common source	Field sales (33%)	Field sales (36%)	Field sales (37%)	Field sales (35%)
Median employees	3	14	71	85
Median sites	1	1	8	1
Median telecom spend now (Rs/mo)	900	2,600	9,150	14,300
Leads with GSTIN	52.8%	86.1%	98.3%	99.3%
Most-bought products (share of activated)	Business Postpaid 299 (100%), Biz Fibre 100 (92%), Cloud POS (34%)	Cloud Email (100%), Postpaid 599 (52%), Endpoint Security (49%)	Postpaid 599 (100%), Backup Failover (57%), Biz Fibre 100 (52%)	Postpaid 599 (100%), Cloud Email (100%), Leased Line 20 (51%)
Median bundle (Rs/mo)	2,470	6,575	**28,080**	15,910
6-month churn	**22.7%**	13.8%	10.8%	8.1%
Mean NPS (0–10)	7.05	6.92	6.70	6.46
Mean days to activate	16.6	16.9	18.8	20.5
Main drop reason	Unreachable (16%)	Unreachable (18%)	Unreachable (20%)	Unreachable (20%)
Cost per activation (Rs)	20,007	**10,695**	15,614	15,324
Typical businesses	Stationery shop, salon, boutique, mobile repair	Law office, coaching centre, travel agency, diagnostic lab	Budget hotel group, electronics chain, pharmacy chain, apparel chain	Plastic moulding, packaging, textile, auto-parts

Source: [F], except mean days, churn and NPS, which come from [S] Q4.



Interpretation: no segment wins on every measure. Manufacturers convert best, multi-site businesses are worth the most per customer, and professional offices are the cheapest to win. Micro is the largest segment but loses on all four bars.

5.2 Answers

1. **Highest conversion:** small manufacturers (12.8%), though from only 289 leads [F].
2. **Greatest revenue opportunity:** multi-site businesses, at a median of Rs 28,080 a month per customer [F]. On first-year value per lead (conversion × median bundle × 12 × (1 − churn)), multi-site also comes first by a wide margin. Micro offers the greatest *volume* (54.2% of leads), but at Rs 2,470 a month.
3. **Lowest acquisition cost:** professional offices (Rs 10,695 per activation) [F].
4. **Highest churn:** micro retail (22.7%) [S]. Within micro, shops currently on a mobile hotspot churn at 29%, against 13% for micro shops that already have broadband [F].
5. **Easiest to onboard digitally:** professional offices (**interpretation**). They usually have one site, 86% have a GSTIN, their core bundle installs in 5 days, and they already live on email and cloud tools [F][B]. Micro is simple too, but only 52.8% have a GSTIN, which blocks the GST-only products [F]. Multi-site and manufacturer bundles involve SD-WAN, leased lines and installs of 14–30 days [B].
6. **Target first:** professional offices, with multi-site run in parallel as an assisted track (5.3).

5.3 The ranking formula and its limits

Phase 1.3 defined the score as **share of base × conversion × median bundle × 1 / churn** [B].

Rank	Segment	Score index (top = 100)	Payback (months of bundle to cover CAC)
1	Small multi-site businesses	100	0.6
2	Professional offices	31	1.6
3	Small manufacturers	29	1.0
4	Micro retail & services	6	8.1

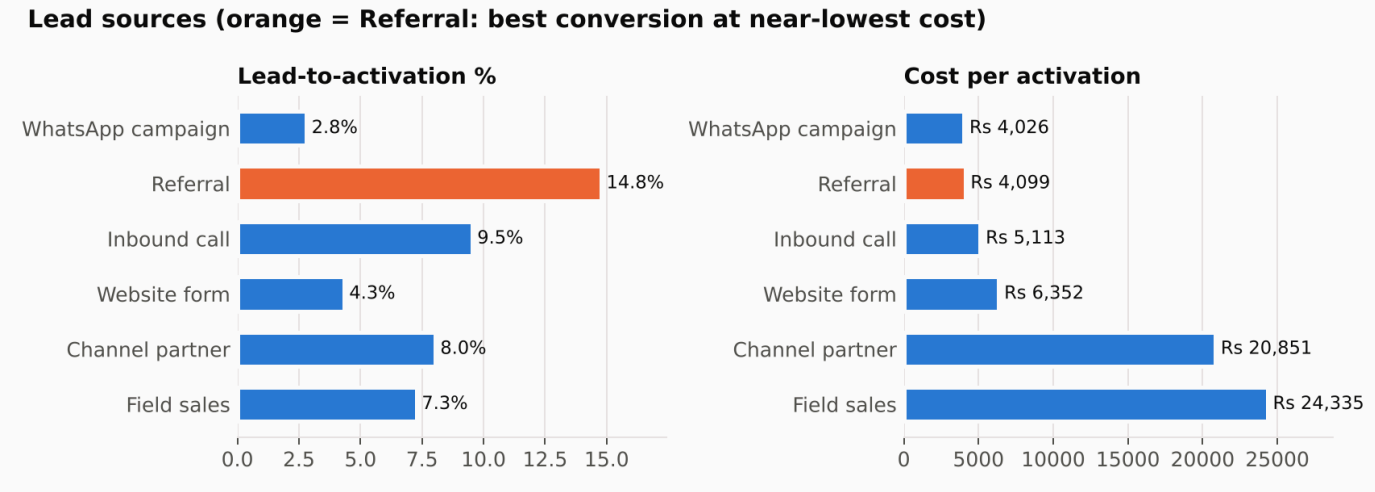
The formula ranks multi-site first. We still recommend launching with professional offices, because the formula leaves out three things:

- **It ignores fit with self-serve.** The case asks for “minimal human effort” [Case]. Multi-site bundles need SD-WAN (minimum 3 sites, 14-day install) and installs at around 7 sites [B]. A human will usually have to close that deal.
- **It ignores what the digital journey changes.** Micro’s score is low *today* because of its Rs 20,007 CAC and 32% quote acceptance [F][S]. Those are exactly the numbers a self-serve journey attacks, so micro is the phase-2 scale target rather than a segment to drop.
- **It treats a small sample as precise.** The manufacturer figures rest on 37 activated customers, so their churn (8.1%) and conversion (12.8%) carry wide uncertainty.

Section 6 — Lead Source and Acquisition Analysis

Source	Leads (share)	Conversion	Cost per lead (Rs)	Cost per activation (Rs)	Never contacted	Churn (activated)	NPS	Median bundle (Rs)	Share of all spend	Share of activations
WhatsApp campaign	358 (7.2%)	2.8%	112	4,026	58.7%	20.0%	7.10	6,915	1%	3%
Referral	413 (8.3%)	**14.8%**	605	4,099	21.1%	24.6%	7.11	5,730	5%	17%
Inbound call	504 (10.1%)	9.5%	487	5,113	19.8%	14.6%	6.65	6,960	4%	13%
Website form	995 (19.9%)	4.3%	274	6,352	45.1%	11.6%	7.07	8,110	5%	12%
Channel partner	1,010 (20.2%)	8.0%	1,672	20,851	28.8%	18.5%	6.75	5,110	30%	22%
Field sales	1,720 (34.4%)	7.3%	1,769	**24,335**	26.6%	9.6%	6.80	7,420	55%	34%

Source: [F]. Blended cost per activation is Rs 15,053 [S].



Interpretation: WhatsApp and referral cost about the same per activation, but referral converts five times better. Field sales and channel partners absorb 85% of spend and deliver 56% of activations.

Highest volume: field sales (34.4% of leads).

Highest-quality customers: it depends on the measure.

- Field sales customers churn least (9.6%) [F].
- Website-form customers have the highest median bundle (Rs 8,110) and low churn (11.6%) [F].
- Referral customers convert best but churn most (24.6%) [F]. Referrals are still the most efficient channel, but they need onboarding care after activation.

Cheapest activations: WhatsApp campaign (Rs 4,026) and referral (Rs 4,099) [F]. WhatsApp gets there by being cheap per lead, not by converting well: only 2.8% of its leads activate.

Greatest digital opportunity: website-form and WhatsApp leads.

- **Evidence.** 45.1% of website-form leads and 58.7% of WhatsApp leads are never contacted at all [F].
- **Evidence.** 34% and 44% of their drops respectively are “unreachable” or “wrong number” [S].
- **Interpretation.** These owners raised their hand online and then waited for a call. An advisor that answers immediately, in the same channel, is the most direct fix in the whole dataset.

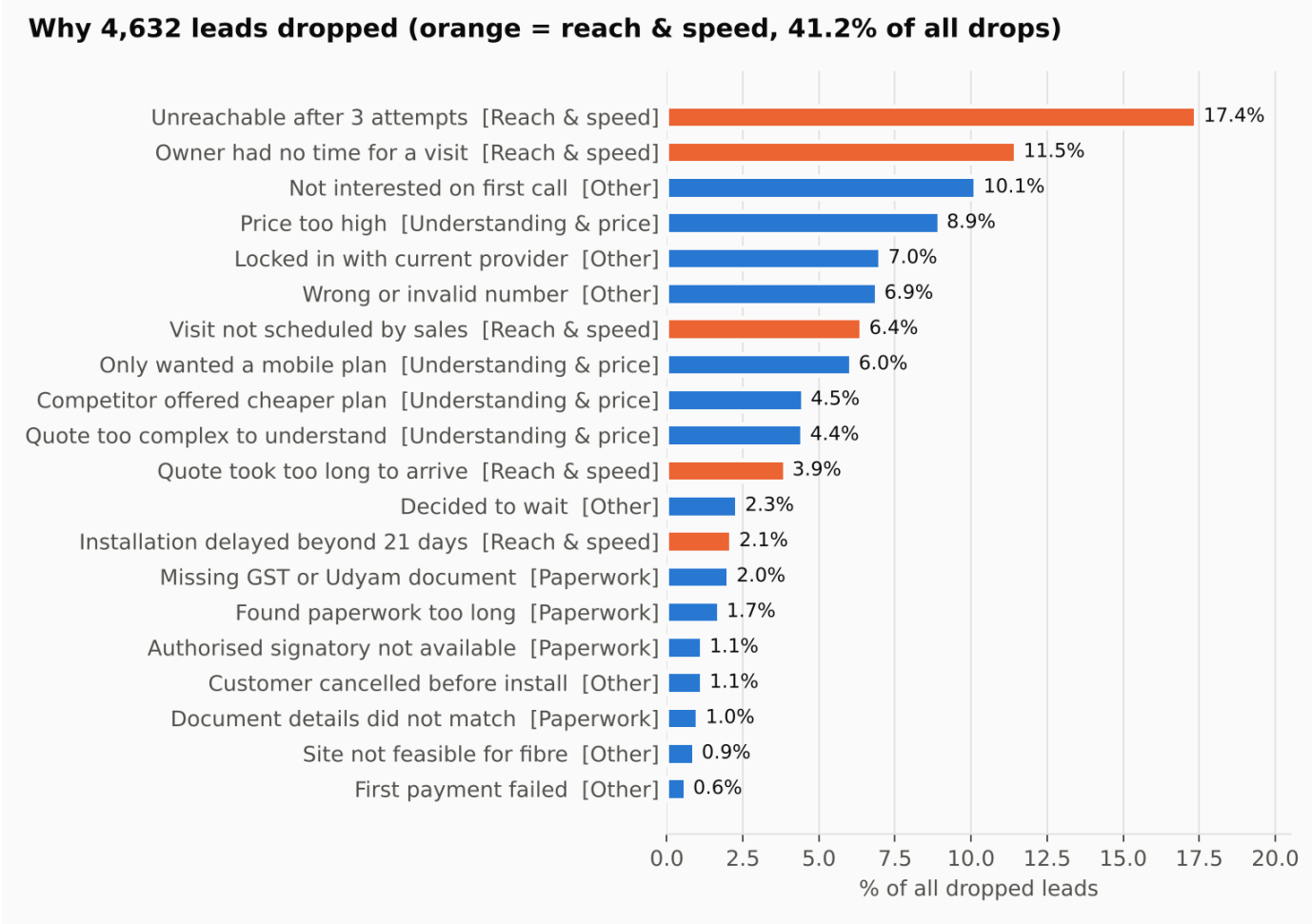
Recommended channel priorities:

1. **Route website and WhatsApp leads straight into the advisor.** They’re cheap, already digital, and today they’re mostly wasted.

2. **Build referral into the product.** Give every activated customer a referral link, and give partners referral codes.
3. **Point field sales and partners at the high-value, assisted cases.** That means multi-site deals, manufacturers and KYC exceptions, with the app doing discovery and the quote first (Section 14.8).

Section 7 — Customer Drop-Off and Root Cause Analysis

7.1 Drop reasons grouped



Interpretation: the two largest reasons, unreachable leads and owners with no time for a visit, are both about reach and speed (orange), and together they cause 28.9% of all drops. Paperwork is real, but it's small in volume.

Group	Leads	% of all 4,632 drops	Most frequent reasons	Business impact	Software can fix?	Human needed?
Reach & speed	1,909	41.2%	Unreachable (805), no time for visit (531), visit not scheduled (296), slow quote (179), install over 21 days (98)	The largest leak; acquisition money spent on leads nobody reaches	**Mostly.** Self-serve removes the visit and the quote wait; the tracker covers installs	For installs
Understanding & price	1,107	23.9%	Price too high (414), only wanted a mobile plan (280), competitor cheaper (207), quote too complex (206)	Lost at the quote, the weakest step	**Partly.** Plain-language quote, right-sized bundles, add-on toggles, mobile-only option	No

Group	Leads	% of all 4,632 drops	Most frequent reasons	Business impact	Software can fix?	Human needed?
Paperwork	273	5.9%	Missing GST/Udyam (94), paperwork too long (79), signatory unavailable (53), details didn’t match (47)	All of the KYC-stage losses	**Mostly.** Registry lookup, OCR pre-fill, mismatch caught at upload, e-sign	For flagged cases
Other	1,343	29.0%	Not interested on first call (470), locked in with current provider (325), wrong number (319), decided to wait (107), cancelled before install (53), fibre not feasible (41), payment failed (28)	Mostly outside product control	**Rarely.** AirFiber fallback for no-fibre sites; payment retry	For retention offers

Source: [S] Q3.

7.2 Root-cause matrix

Root cause	Recorded symptoms	Software fix	Human role
Discovery needs a visit	No time for visit; visit not scheduled	Advisor chat or voice, available any time	None for the standard case
Leads wait for an outbound call	Unreachable; wrong number	The owner starts the session; WhatsApp follow-up	Call High-propensity leads first (Section 11)
Quotes made by hand, with product codes	Slow quote; quote too complex	Instant quote in plain language	None
Bundle not tailored	Price too high; only wanted mobile	Rules-checked bundle sized to the business, add-on toggles	None
Manual documents	Missing doc; paperwork too long; mismatch	Lookups, OCR, cross-checks at upload	Review flagged cases
No status visibility	3.0 chase calls; churn rising with chasing	Live tracker and proactive updates	Named agent when a step runs late
Physical install capacity	Install over 21 days; not feasible	Expected dates; AirFiber fallback	Field installation teams

7.3 How much a digital journey can realistically address

Each of the 20 drop reasons was classified as directly, partly or not addressable. The reasoning for each one is in [S] Q3.

Class	Reasons	Leads	% of all drops
Directly addressable	No time for visit; visit not scheduled; slow quote; complex quote; missing GST/Udyam; paperwork too long; details didn’t match; fibre not feasible; payment failed	1,501	**32.4%**
Partially addressable	Unreachable; install delayed; price too high; only wanted mobile; signatory unavailable; cancelled before install	1,703	**36.8%**

Class	Reasons	Leads	% of all drops
Needs offline intervention or is out of reach	Not interested; locked in; wrong number; competitor cheaper; decided to wait	1,428	**30.8%**

Source: [S].

Assumption. The “partly” class won’t be fully recovered. Read **32% as the floor** and **69% as the ceiling** of drops the journey can influence. Even recovering a fraction of the floor would lift conversion meaningfully (Section 16).

Section 8 — Bundle and Product Intelligence

8.1 What the catalogue allows

Source for this subsection: [F].

- **25 products:** Connectivity 11, Communication 4, Cloud & Productivity 4, Security 3, IoT & Physical 3.
- **10 need an active GSTIN:** P03, P05, P06, P10, P12, P13, P14, P15, P21, P24.
- **7 need a base product in the same bundle:** P09, P10, P11, P12, P21, P22, P23.
- **One is sold in tier 1 cities only:** P03 Biz Fibre 1G.
- **Minimum employees:** P03 (5), P05 (10), P06 (25), P10 (10), P12 (3), P21 (10), P24 (10).
- **Minimum sites:** P10 Managed SD-WAN (3).
- **Setup, contracts and installs:** 12 products have a setup fee. Contracts run 1–24 months, and installs take 1–30 days.

Traps worth knowing:

- **Static IP (P11)** needs Biz Fibre (P01–P03). It can't go with a leased line, even though leased-line customers are the most likely to want one [S].
- **WhatsApp Business API (P15) and Business SMS (P14)** need a GSTIN. They are among the most-asked-about products (16.5% and 16.3% of leads), and 47.2% of micro leads have no GSTIN [F].

8.2 What customers actually buy

Most purchased (share of 368 activated customers) [F]:

- Business Postpaid 599 52%
- Biz Fibre 100 49%
- Business Postpaid 299 48%
- Cloud Email & Office Suite 48%
- Biz Fibre 300 27%
- Cloud CCTV 23%
- Endpoint Security 23%

Bought together. The most frequent pair is Biz Fibre 100 + Business Postpaid 299, in 116 baskets (31.5%). The strongest attachments by lift are [S]:

Pair	Baskets	Lift
Cloud Telephony + Business SMS	14	5.02
Managed SD-WAN + Toll-free Number	12	4.15
Biz Fibre 1G + Static IP	20	2.95
Cloud Backup + Endpoint Security	32	2.53

Popular combinations by segment [S]:

- **Micro** has a clear default: Biz Fibre 100 + Business Postpaid 299 (21 of 110 buys).
- **The other segments** buy almost-unique bundles: professional offices 138 customers, manufacturers 37, multi-site 83, with few repeats.
- **Interpretation.** A conversational advisor is a better fit than a fixed menu.

Value, setup and contract [F]:

- Median bundle Rs 6,900 a month.
- Setup fee: median Rs 2,650, mean Rs 13,879. The mean is pulled up by multi-site and manufacturer installs.
- Contracts: 304 of 368 customers are on 12 months, 61 on 24 months and 3 on 6 months.

Bundle size and churn [S]:

- 2-product bundles churn at 24.0%.
- 3–6 product bundles churn at 10.4–15.9%.
- 7+ product bundles churn at 20.6%.
- **Interpretation.** Very thin and very large bundles both carry more risk.

8.3 Recommended starter bundles (all validated)

Method [B]:

- **Core** = products at least 40% of the segment's activated customers bought.
- **Add-ons** = products 25–40% bought.
- **Quantities:**
  - per-site items = number of sites;
  - per-user items = the smaller of the segment median and the headcount;
  - everything else = the segment median.
- **Pricing:** micro at the catalogue minimum (very price-sensitive); other segments at the middle of the range.

`analysis/validate_bundles.py` ran **22 bundle checks and 9 must-refuse checks, with 0 failures** [B].

Bundle	Profile tested	Core	Monthly	Setup	Contract	Install	With add-ons
Dukaan Digital (micro)	3 staff, 1 site; also 2 staff, tier 2, no GSTIN	Biz Fibre 100 + 2× Postpaid 299	Rs 1,597	Rs 0	12 mo	5 days	Rs 2,794 (POS, Wi-Fi, CCTV)
Office Pro (professional)	13 staff, 1 site, tier 2	Biz Fibre 300 + 3× Postpaid 599 + 7× Cloud Email + 6× Endpoint + Backup	Rs 6,055	Rs 0	12 mo	5 days	Rs 10,055 (3× Cloud Telephony, Static IP)
Chain Connect (multi-site)	7 sites, 61 staff, tier 2	7× Biz Fibre 100 + 7× Failover + 7× Cloud POS + 4× Postpaid 599	Rs 19,575	Rs 28,000	12 mo	5 days	Rs 49,557, 24 mo (SD-WAN, CCTV, Wi-Fi, Toll-free, WhatsApp)
Factory Line (manufacturer)	74 staff, 1 site, tier 2	Leased Line 20 + 4× Postpaid 599 + 11× Cloud Email + 9× Endpoint	Rs 15,430	Rs 10,000	12 mo	21 days	Rs 21,164, 24 mo (Firewall, CCTV, 10× Machine IoT)

Why each bundle makes commercial sense (**interpretation**, using numbers from [B][F]):

- **Dukaan Digital** is priced within reach of what micro customers pay today (median Rs 2,470 for activated bundles). It works without a GSTIN. The GST-only WhatsApp add-on appears only after the GSTIN lookup passes, and AirFiber (Rs 1,497, 2-day install) is the fallback where fibre isn't feasible.
- **Office Pro** at Rs 6,055 sits just under the segment's Rs 6,575 median. The 5-day install keeps activation short. A 4-person clinic gets Rs 5,393, because per-user items scale with headcount. The tier-1 upgrade to 1G fibre is correctly blocked for teams under 5 people.
- **Chain Connect** core comes in below the Rs 28,080 median. SD-WAN is the natural add-on because the segment asks for it most (14% of multi-site leads) and pairs it with Toll-free (lift 4.15) [S][F]. Its 14-day install is why this segment should be an assisted track.
- **Factory Line** matches the Rs 15,910 median. The leased line drives a 21-day install. Static IP is deliberately left out because it can't go with a leased line.

Must-refuse cases the advisor has to handle [B]:

- WhatsApp API without a GSTIN
- 1G fibre in a tier 2 city
- CCTV with no connection
- SD-WAN for 2 sites
- Leased line on a cancelled GSTIN

- Static IP on a leased line
- Cloud Telephony for 2 people
- Machine IoT for 6 people
- Toll-free on a suspended GSTIN

Section 9 — KYC and Document Verification Findings

9.1 What was processed

All **47 images** for applicants K01–K12 were run through `scripts/kyc_check.py` [K]. The pipeline:

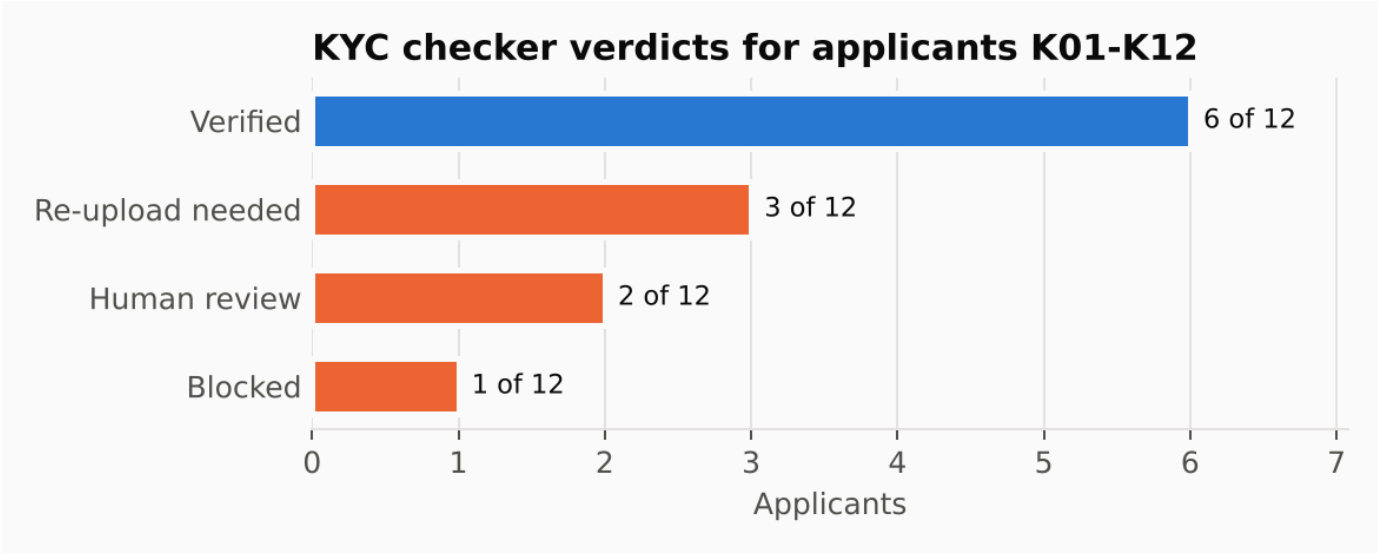
1. **Clean the image.** Strip the pink “SYNTHETIC SPECIMEN” watermark by colour, then threshold.
2. **Fix problem images.** Straighten tilted scans; measure sharpness and flag blurry images.
3. **Read the text** with Tesseract OCR.
4. **Repair the GSTIN.** Fix look-alike characters (0/O, 1/I, 2/Z, 5/S, 3/9, I/J, and so on) and keep only a candidate that passes the official GSTIN checksum *and* exists in the registry.
5. **Run every cross-check:**
    - registry status
    - legal and trade name
    - state code
    - Udyam number, name, city and state
    - bill date
    - bill name and address
    - signatory business, expiry and proprietor match
    - missing documents

Every result below comes from actually running that pipeline. The visual reading of each GSTIN was also checked against the checksum and the registry to confirm the pipeline’s verdicts.

9.2 Registry and OCR facts

- **OCR quality:** median confidence across the 47 documents is 91.4%. The lowest is 43.8%, K04’s blurred GST certificate [K].
- **GSTIN is the hardest field.** It’s printed in a monospace font and the watermark crosses it. Raw OCR read only 1 of 11 readable GSTINs exactly right (K05). The checksum-plus-registry repair recovered 9 more automatically. The remaining one, K06, is a genuine problem and not an OCR error [K].
- **K02’s GST certificate is tilted.** It was straightened by 7° automatically and then verified [K].
- **Name-only lookups are unsafe.** “Vindhya Tours & Travels” (K01) has Udyam records in Maharashtra, Delhi and Telangana (Hyderabad), but none in Visakhapatnam where K01 is. Matching on name alone would attach the wrong business [K].

9.3 Verdicts



Interpretation: half the sample is clean, so the happy path matters. The other half spreads across three outcomes that need different treatment: re-upload, human review, and hard block.

Applicant	Business	Verdict	Planted issue found [K]
K01	Vindhya Tours & Travels	**Blocked**	GSTIN 37IHKXV5167L1ZS is **Cancelled** in the registry; Udyam number field is **blank**; no Udyam record for this name in Visakhapatnam
K02	Harmony Dental Care	Verified	None (tilted scan straightened automatically)
K03	Ganesh Woodcraft	Verified	None
K04	Saffron Dental Care LLP	**Re-upload**	GST certificate **too blurry to read** (sharpness 1.0, OCR 43.8%). The registry lookup by name shows the GSTIN is Active
K05	Balaji Realty LLP	**Re-upload**	**Udyam certificate missing** (11 Udyam certificates for 12 applicants)
K06	Saffron Unisex Salon	**Human review**	The GSTIN printed on the certificate **fails the checksum** and differs from the registry GSTIN for this business at character 8 ('8' printed, '5' in the registry). Not an OCR look-alike
K07	Neelgiri Physio Clinic	**Human review**	Legal name reads "Neha Banerjee (Prop. **Shri** Physio Clinic)", but the trade name and registry say **Neelgiri** Physio Clinic
K08	Green Leaf Legal Associates	Verified	None
K09	Kaveri Insurance Advisors	Verified	None
K10	Kaveri Stays LLP	Verified	None (signatory authority note, see 9.4)
K11	Trinity Bake House Pvt Ltd	**Re-upload**	Address proof dated **2025-10-31**, which is 257 days before 2026-07-15 (limit 90)
K12	Star Electronics	Verified	None

Limitation. The judges hold the official KYC answer key [Case]. Our checker finds 6 flagged applicants. The key may list issues our checks don't look for, so compare against it as soon as it's available.

9.4 Handling each issue type

Issue	What is wrong	How it's detected	What the customer is told	Auto-reject?	Human review?
Cancelled or suspended GSTIN (K01)	The business can't legally be sold GST-only products	Registry `status` ≠ Active	"Your GST registration shows as cancelled. You can still buy plans that don't need GST, or contact us if this is wrong."	**Yes** for GST-only products; offer non-GST bundle	Only if the customer disputes it
Blank Udyam number (K01)	The certificate is incomplete	Regex finds no number in the field	"We couldn't find a registration number on your Udyam certificate. Please type it or upload a clearer copy."	No	If still unresolved
Blurry image (K04)	Document can't be read	Sharpness score below threshold, low OCR confidence	"This photo is blurry. Please retake it in good light, holding the phone flat."	No: ask for a re-upload on the spot	No

Issue	What is wrong	How it’s detected	What the customer is told	Auto-reject?	Human review?
Missing document (K05)	Required document not uploaded	Document checklist	“We still need your Udyam certificate. You can upload it or enter the number.”	No	No
GSTIN on certificate ≠ registry (K06)	Possible typo, tampering or wrong certificate	Checksum fails, and the difference isn’t an OCR look-alike	“The GST number on your certificate doesn’t match government records. A specialist will check it with you.”	**No:** never auto-reject a possible fraud flag	**Yes**
Legal name names a different business (K07)	Certificate inconsistent with registry	Word-level name match against trade name and registry legal name	“Your GST certificate shows a different business name. A specialist will confirm the details.”	No	**Yes**
Stale address proof (K11)	Bill older than 90 days	Bill date vs 2026-07-15	“Your bill is from Oct 2025. Please upload one from the last 3 months.”	No: ask for a re-upload	No
Tilted scan (K02)	OCR accuracy drops	Page-angle detection	Nothing; it’s fixed silently	No	No
Company/LLP signatory (K04, K05, K10, K11)	Registry has no director list, so signing authority can’t be proven	Constitution ≠ Proprietorship	“Please confirm you’re authorised to sign for the company (e-sign or authorisation letter).”	No	Spot-check

Recommendation. Use three outcomes:

- **Fix it now:** re-upload or type the number.
- **Specialist review:** mismatches and possible fraud.
- **Hard block:** a cancelled GSTIN for GST-only products.

Only a cancelled or suspended GSTIN should block anything automatically. Everything else gives the owner a clear next step.

Section 10 — Customer Experience Insights

10.1 What the lead data says about customers

Source for this subsection: [F], unless marked [S].

Languages.

Language	Share of leads	Conversion
Hindi	42.7%	7.9%
English	23.4%	8.0%
Marathi	10.1%	6.3%
Kannada	5.1%	5.1%
Gujarati	4.7%	6.0%
Telugu	4.1%	7.7%
Tamil	4.1%	4.3%
Bengali	2.2%	9.0%
Punjabi	1.9%	4.2%
Malayalam	1.7%	8.4%

- Hindi and English cover 66.1% of leads, and regional languages the other 33.9%.
- In each state, the local language is the top choice for 42–52% of leads (e.g. Tamil 45% in Tamil Nadu, Marathi 49% in Maharashtra).
- Regional-language leads accept the quote 40% of the time, against 45% for Hindi and English [S].

Business types. Ten micro types each have 258–291 leads and convert at 3.0–5.7%. Among professional offices, law offices convert at 14.1% and coaching centres at 8.9%.

Size. Median 6 employees; the middle half has 3–21. A quarter of leads have 3 or fewer.

Current internet.

Current setup	Leads	Median spend (Rs/mo)	Conversion
Mobile hotspot	1,799	950	4.5%
Business broadband	1,433	3,250	9.3%
Home broadband	1,250	1,300	7.6%
Mixed per site	221	9,300	12.7%
Leased line	173	11,750	15.0%

Current provider.

- **1,890 leads (37.8%) already use Accel**: 1,505 for mobile only and 385 for more. They convert at 9.2%, against 6.2% for everyone else.
- Competitor A converts at 6.5%, Competitor B at 7.4% and local ISPs at 5.4%.

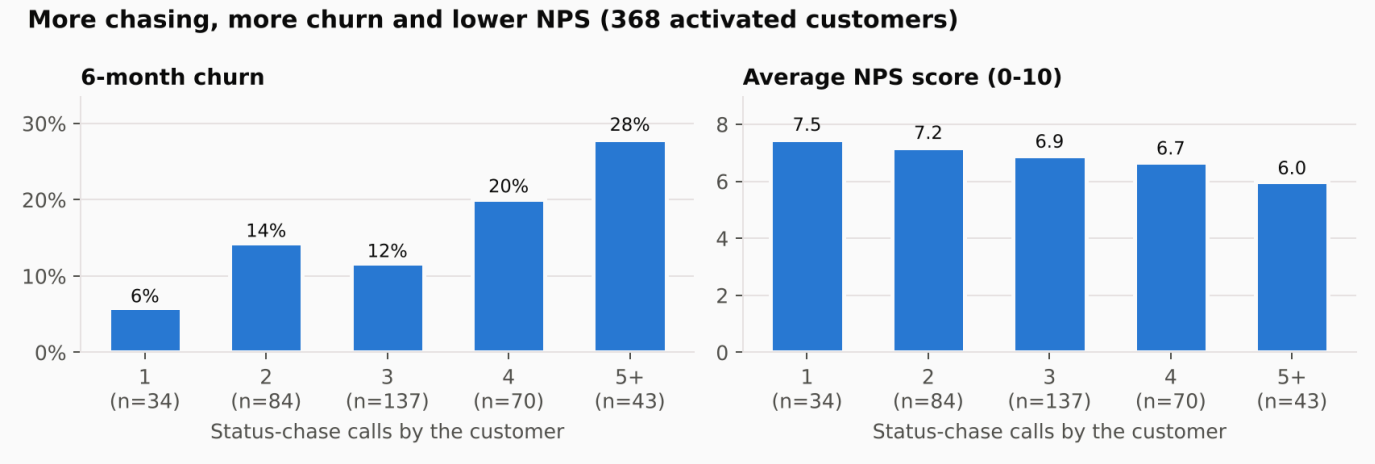
Telecom spend.

- Median by segment: micro Rs 900, professional Rs 2,600, multi-site Rs 9,150, manufacturer Rs 14,300 a month.
- Activated customers' bundles cost a median **2.59×** what they spent before. By segment: micro 2.74×, professional 2.66×, manufacturers 1.23×, multi-site 3.04×.

Products of interest.

- The most-asked-about products are Postpaid 299 (16.8% of leads), WhatsApp API (16.5%), Secure Wi-Fi (16.3%), Business SMS (16.3%), Cloud CCTV (15.7%) and Website (15.0%).
- Leads asking about those convert at only 4.5–6.4%.
- Leads asking about Cloud Backup (14.1%), 1G fibre (12.5%), Failover (12.5%), Firewall (12.4%) or Endpoint Security (12.2%) convert at roughly double that rate.

Follow-up behaviour. 68% of activated customers chased status 3 or more times. Chasing rises churn from 6% to 28% and lowers NPS from 7.5 to 6.0 [S].



Interpretation: both panels move in the same direction as chasing increases. Chase calls barely track how long activation took (correlation +0.01) [S], so it's the not-knowing that hurts, not just the wait. A status page the owner can check is the remedy.

Activation duration. Mean 17.6 days, range 7–31. 23% activated within 14 days and 16% took over 21 days.

NPS. Among activated customers, 18% are promoters, 38% passives and 43% detractors, for an NPS-style score of −25. The mean score is 6.86.

10.2 Personas, built from the actual slices

All figures are from [F].

1. Ravi, mobile-hotspot shopkeeper (micro on hotspot: 1,597 leads, 31.9%)

- **Profile:** median 4 staff, 1 site, spends Rs 900/month. 53% have a GSTIN. Hindi 44%, English 24%, Marathi 10%. Mobile repair shops, stationery shops, bakeries, pharmacies. 41% of contacts are the owner.
- **Main pain:** unreachable (18%), no time for a visit (12%), price too high (11%). Converts at 4.1%, churns at 29% after activation.
- **What he asks for:** WhatsApp API (23%), website (23%), guest Wi-Fi (22%), a business SIM (22%).
- **Best channel:** voice or WhatsApp, in Hindi or a regional language, outside shop rush hours.
- **Likely objections:** "Too expensive." "I don't have GST." "I can't leave the shop."
- **Recommended experience:** a 3–4 question voice chat and a Rs 1,597 core bundle. WhatsApp API is shown only after a GSTIN passes. AirFiber is offered if there's no fibre. A clear monthly total with nothing hidden.

2. Meena, shop owner who already has broadband (micro with broadband: 1,112 leads, 22.2%)

- **Profile:** median 3 staff, Rs 900/month. Salons, boutiques, sweet shops.
- **Main pain:** unreachable (14%), no time for a visit (13%).
- **What she asks for:** Cloud CCTV (28%), AirFiber (23%), Wi-Fi (22%), POS (22%).
- **Outcome after activation:** churn 13%, much lower than Ravi's 29%.
- **Recommended experience:** lead with the add-on she asked about, CCTV or POS, on top of a fibre-plus-SIM core.

3. Dr Arjun, professional office (1,247 leads, 24.9%)

- **Profile:** median 14 staff, 1 site, Rs 2,600/month. 86% have a GSTIN. Law offices, coaching centres, travel agencies, diagnostic labs. 40% of contacts are the owner, 20% a partner.
- **Main pain:** unreachable (18%), not interested on first call (10%), no time (10%).
- **Asks about:** Biz Fibre 300 (13%), Endpoint Security (12%), Toll-free (12%).
- **Outcome after activation:** median bundle Rs 6,575, churn 14%, 16.9 days to activate.
- **Best channel:** chat on a phone or laptop, English or Hindi, any time.
- **Likely objections:** data security, downtime, "will email migration be painful?"
- **Recommended experience:** Office Pro with seats auto-sized to headcount, uptime and security explained in plain words, a 5-day install date shown upfront.

4. Priya, multi-site chain operator (755 leads, 15.1%)

- **Profile:** median 71 staff, 8 sites, Rs 9,150/month. 98% have a GSTIN. Hotel groups, electronics, pharmacy and apparel chains.
- **Asks about:** SD-WAN (14%), WhatsApp API (13%), SMS (12%).
- **Main pain:** unreachable (20%).
- **Outcome after activation:** median bundle Rs 28,080, churn 11%, 18.8 days to activate.
- **Recommended experience:** the app builds a per-site quote and books a survey. A named account manager closes the deal.

5. Suresh, small manufacturer (289 leads, 5.8%)

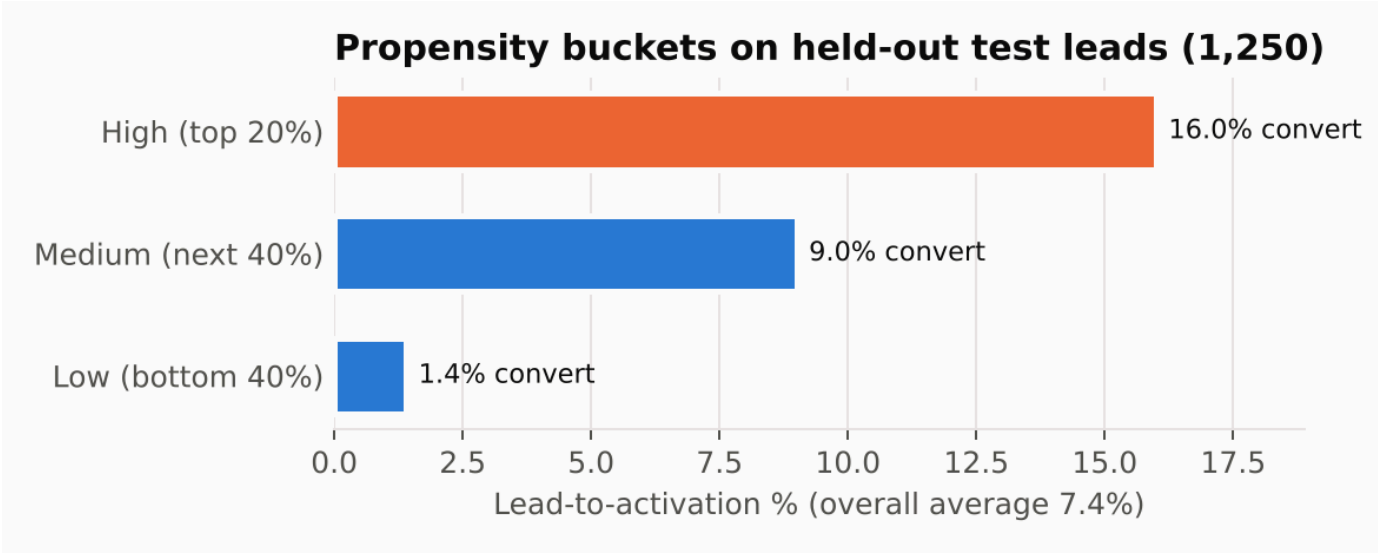
- **Profile:** median 85 staff, 1 site, Rs 14,300/month. 99% have a GSTIN. Moulding, packaging, textile and auto-parts units.
- **Asks about:** Cloud Backup (20%), Postpaid 599 (18%), Endpoint Security (17%), Static IP (17%).
- **Outcome after activation:** median bundle Rs 15,910, churn 8%, 20.5 days to activate.
- **Recommended experience:** the advisor explains why Static IP needs fibre, offers the leased-line or 1G route, and hands over to a specialist for the site survey.

10.3 What this means for the UI and the advisor

- **Languages.** At launch, support Hindi and English plus the five largest regional languages: Marathi, Kannada, Gujarati, Telugu and Tamil. Together that’s 94.2% of leads [F]. Offer voice for owners who’d rather speak than type.
- **Questions.** Ask at most 5–6 short questions. Ask about the GSTIN early, because it decides what can be sold, and a lead with a GSTIN is 4.7× more likely to activate [S].
- **Pricing language.** Show the monthly total, then a “compared with what you pay now” line. Bundles cost a median 2.6× previous spend [F], so explain what’s new (CCTV, POS, backup) rather than hiding the total.
- **Ineligible requests.** When an owner asks for something they can’t have yet (e.g. WhatsApp API without a GSTIN), say why and what unlocks it. Don’t silently drop it.
- **Status updates.** Make status the home screen after purchase, with proactive WhatsApp updates, so the owner never needs to chase.

Section 11 — Propensity Model Findings

Item	Detail [M]
Target	`outcome == "Activated"` (7.4% of leads)
Features	Segment, lead source, preferred language, current internet, turnover band, has GSTIN, has Udyam, city tier, employees (log), sites, telecom spend (log), number of products asked about. All known when the lead is created
Left out on purpose	Stage dates, drop reason, chase calls, bundle and value fields: these are only known later, so including them would be cheating ("leakage"). Also GSTIN *status*: past sellers closed deals on cancelled GSTINs, so the model would have learned to reward a compliance problem
Preparation	One-hot encoding for categories, scaling for numbers, class balancing because only 7.4% are positive
Method	Stratified 75/25 split (3,750 train / 1,250 test) plus 5-fold cross-validation
Models	Logistic regression: test AUC **0.731**, cross-validated AUC 0.706. LightGBM: test AUC 0.688
Chosen	Logistic regression. It's at least as accurate, and every reason can be explained
Precision in top 20%	The top-scored 20% convert at **16.0%**, against 7.4% overall, and hold **43%** of all activations



Interpretation: the three buckets separate cleanly on leads the model never saw. Low-bucket leads almost never convert (1.4%).

Buckets.

- On the test set: High 16.0%, Medium 9.0%, Low 1.4%.
- After refitting on all 5,000 leads: High 15.9%, Medium 8.7%, Low 1.8% [M].

Top factors.

- **Raise the odds:** came via referral, has a GSTIN, came via inbound call, manufacturer or multi-site segment, prefers Hindi or English, on a leased line or mixed setup, turnover Rs 5–25 crore.
- **Lower the odds:** WhatsApp-campaign source, micro segment, no GSTIN, website-form source, prefers Tamil or Kannada, tier 2 city, turnover under Rs 40 lakh.

How admins use it.

- The dashboard sorts unclosed leads by bucket and shows the top 3 reasons ("+ came via Referral, + has a GSTIN, − tier 2 city").

- Calling only the High bucket reaches 43% of future customers with a fifth of the calls [M].
- Low-bucket leads stay in the self-serve journey and get WhatsApp nudges, not calls.

Correlation is not causation.

- **Interpretation.** “Prefers Tamil” lowers the score. That doesn’t mean Tamil speakers want the products less; it more likely reflects how badly today’s process serves them.
- **Interpretation.** WhatsApp-campaign leads score low because nobody called them back (58.7% were never contacted [F]), not because WhatsApp brings poor customers.
- **Recommendation.** The model tells us *whom to call first today*. It must not decide whom to serve, and we should expect its weights to change once the journey changes.

Limitations.

- An AUC of 0.73 is useful but modest.
- Scores rank leads; they aren’t calibrated probabilities. Show the bucket’s conversion rate, not the raw score.
- It’s trained on a synthetic quarter under the old process.
- Retrain it once app data exists.

Section 12 — Business Opportunities

Identified problem	Evidence	Proposed opportunity	Expected benefit	Priority
Discovery needs a visit; leads go unreached	No time for visit 531, visit not scheduled 296, unreachable 805 [S]	**AI business advisor** (chat, then voice) that the owner starts any time	Removes the visit; serves digital leads instantly	Must
Quote is the weakest step	43.6% acceptance; complex 206, slow 179 [F][S]	**Guided package recommendation with instant, plain-language pricing** and add-on toggles	Higher quote acceptance; no 3–5 day wait	Must
Mis-selling	15% of activated bundles break rules [S]; 15 live advisor tests [Case]	**Deterministic rules engine** behind the AI	Zero ineligible bundles; passes judge scenarios	Must
Paperwork drops and late mismatches	273 KYC-stage drops; 47 mismatches [F]	**Digital KYC:** consent, GSTIN/Udyam lookup, pre-fill	Fewer missing documents; faster KYC	Must
Bad documents found too late	6 of 12 applicants flagged [K]	**OCR discrepancy detection** at upload, with specific fix-it messages	Problems fixed in the same session	Must
Customers chase status	3.0 calls; churn 6% → 28% [S]	**Order tracker** with expected dates from catalogue install days	Fewer calls; lower churn	Must
Cases software can’t settle	K06, K07; late installs 98 [K][F]	**Human escalation** with full context and a named agent	Exceptions resolved without the owner repeating themselves	Must
Limited sales capacity spread evenly	Top 20% of leads hold 43% of activations [M]	**Lead prioritisation** (propensity) on the admin dashboard	More activations per call	Should
Regional speakers convert worse	Regional quote acceptance 40% vs 45% [S]; Tamil 4.3%, Punjabi 4.2% [F]	**Multilingual voice interaction**	Narrows the language gap	Should
Digital leads abandon and go silent	WhatsApp 44% / website 34% of drops unreachable [S]	**WhatsApp reminders** (opt-in) for incomplete KYC and status	Recovers stalled journeys	Should
No visibility of whether it works	Baselines exist; no live tracking	**Admin analytics** on the 5 metrics	Shows impact; steers the pilot	Should
Price objections	Price too high 414; competitor cheaper 207 [S]	Price **transparency and right-sizing** (headcount-based seats, catalogue-minimum pricing for micro)	Fewer price drops	Should
Price objections (aggressive option)	Same	**Dynamic pricing / discounts**	Unclear; risks margin and trust	Won’t (now)
Existing Accel customers convert better	37.8% of leads already use Accel; 9.2% vs 6.2% conversion [F]	Pre-filled upgrade journey for existing mobile customers	Cheaper, warmer acquisition	Could
Referral is the best channel	14.8% conversion, Rs 4,099 per activation [S]	Referral link after activation; partner codes	More of the best-value leads	Could

Not recommended now: dynamic pricing. Nothing in the data shows that discounts move conversion, and changing prices per customer would undercut the “one transparent price” the case asks for [Case].

Section 13 — Alternative Product Solutions

A. AI-first self-service onboarding platform

- **Target customer:** every segment, fully self-serve.
- **Main functionality:** advisor, quote, KYC, payment and tracker, with no humans in the loop. Exceptions get an error message or a “call us” number.
- **Journey:** chat → bundle → pay → upload → wait.
- **Technical complexity:** medium.
- **Business value:** high CAC savings on simple cases.
- **Implementation effort:** medium.
- **Risks:**
  - Exceptions dead-end. 6 of 12 KYC applicants needed a fix or a review [K].
  - The 220 post-KYC losses are all physical or payment issues [F].
  - Multi-site deals can’t close without a survey.
  - It contradicts the case’s question 4 on the role of the field force and partners [Case].
- **Advantages:** simplest story; cheapest per activation.
- **Disadvantages:** abandons the hardest and most valuable customers. A dead-end at KYC recreates today’s paperwork drop.

B. Hybrid AI + human-assisted onboarding platform

- **Target customer:** professional offices and micro self-serve; multi-site and manufacturers assisted.
- **Main functionality:** everything in A, plus automatic escalation (late step, KYC flag, advisor stuck, “talk to a person”) to a named agent who sees the full context, plus an admin view with propensity-ranked unclosed leads.
- **Journey:** chat → bundle → consent → KYC → tracker. A human steps in only on exceptions.
- **Technical complexity:** medium-high. It adds an escalation queue, agent assignment and an admin dashboard.
- **Business value:** keeps A’s savings on the common case and recovers the exceptions.
- **Implementation effort:** high, but it can be built in layers.
- **Risks:** scope creep; agent capacity during a pilot.
- **Advantages:**
  - Matches the principle “AI handles acquisition; humans handle exceptions.”
  - Answers every key question in the case.
  - Uses the field force where the data says they add value.
- **Disadvantages:** more to build and demo; needs clear escalation rules.

C. Assisted digital sales platform for agents and channel partners

- **Target customer:** sales staff and partners, who use the app on behalf of owners.
- **Main functionality:** a quote builder, KYC capture and lead scoring for agents.
- **Journey:** agent visits → app builds the quote → agent collects documents.
- **Technical complexity:** medium.
- **Business value:** faster quotes and fewer mis-sells.
- **Implementation effort:** medium.
- **Risks:** keeps the visit, which today causes 827 drops (no time 531, not scheduled 296) [S]. Keeps the Rs 20,851–24,335 per-activation cost of partner and field channels [F].
- **Advantages:** easy adoption by the existing sales force.
- **Disadvantages:** doesn’t meet the case’s core ask: “on a phone, in their preferred language, with minimal human effort” [Case].

Weighted decision matrix

The weights and scores below are the **team’s judgement**, informed by the evidence cited above. They are not computed from the data. Scores run from 1 to 5, and the weighted total is out of 5.

Criterion	Weight	A: AI-first	B: Hybrid	C: Agent tool
Fit to case requirements (self-serve, phone, language, field-force role)	25%	4	5	2
Addresses the observed drop causes (reach/speed 41%, quote 44% pass, KYC 273, install 220)	25%	3	5	3
Feasible within the hackathon	20%	4	3	4
Business value (CAC, conversion, churn)	15%	4	5	3
Risk (dead-ends, compliance, adoption)	15%	2	4	3
Weighted total	100%	**3.45**	**4.45**	**2.95**

Recommendation: B, the hybrid.

- **It’s the only option that covers both ends of the funnel.** That means the 65% of losses before the quote and the exceptions after it (KYC flags, installs) [F][K].
- **A loses** on risk and drop coverage. With no human, the exception path dead-ends, and that path held half the KYC sample.
- **C loses** on fit. It keeps the visit and the most expensive channels, which are the very things the data identifies as the problem.

Section 14 — Recommended Product Strategy

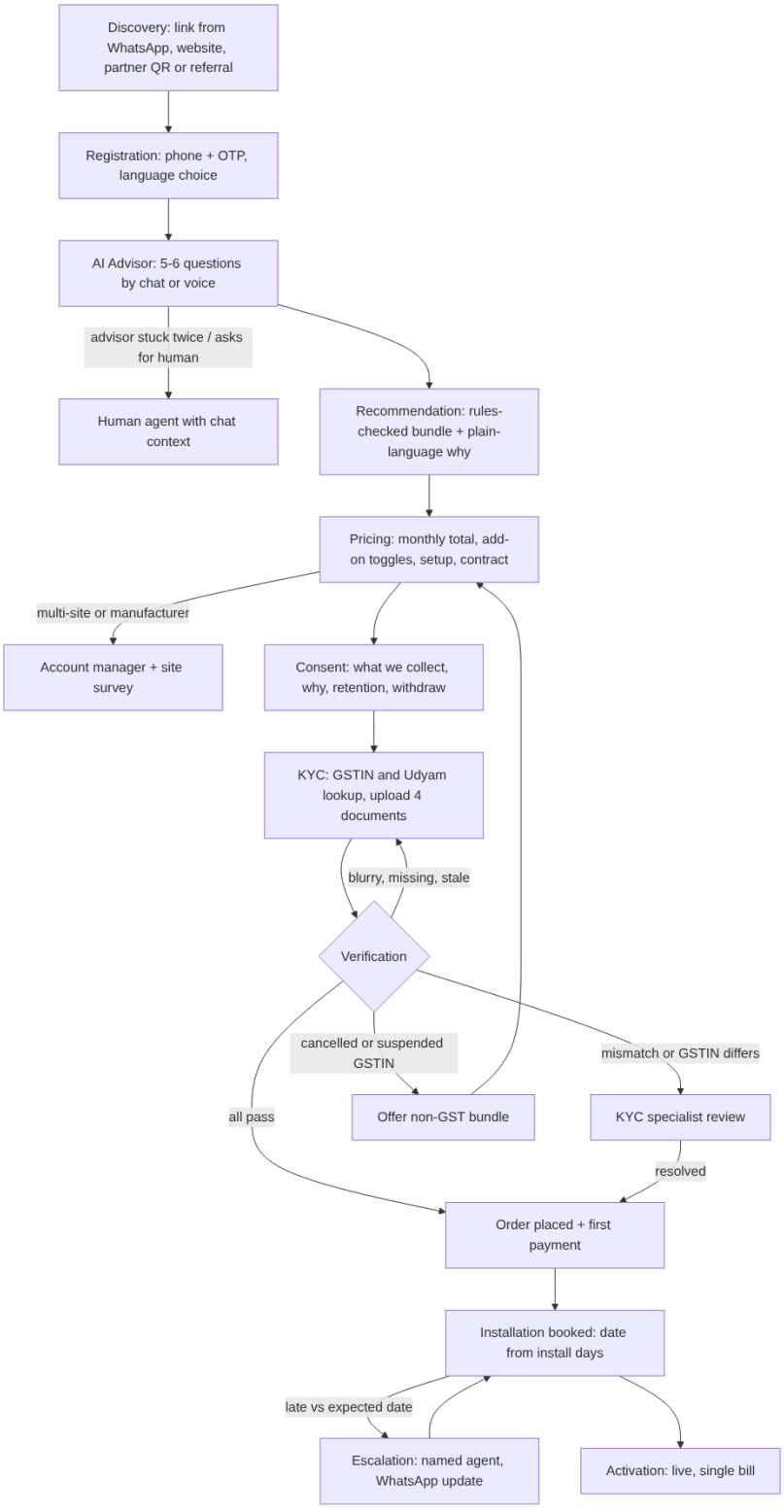
1. **Positioning.** “Your business connected in one sitting.” A phone-first advisor that recommends, prices and verifies in the owner’s language, with a real person when needed.
2. **Primary segment.** Professional offices: clinics, law and CA offices, coaching centres, diagnostic labs. They convert at 11.1%, have the lowest CAC (Rs 10,695), have 86% GSTIN coverage, and a 5-day core install [F][B].
3. **Secondary segments.**
    - **Micro retail (phase 2, scale).** The largest segment (54.2%), and the one whose specific failures the journey fixes: quote acceptance of 32% and CAC of Rs 20,007 [S][F].
    - **Multi-site (assisted track from day 1).** The highest value (Rs 28,080 median), closed by an account manager with the app’s quote [F].
    - **Manufacturers (assisted).** Leased-line installs take 21 days [B].
4. **Core customer problem.** Buying business connectivity today takes a visit, a slow and confusing quote, paper KYC and 17.6 days of waiting, with 3 chase calls [S].
5. **Value proposition.**
    - The right bundle in minutes, priced in plain words.
    - Documents checked while you upload them.
    - A live date you can check instead of calling.
6. **Essential features:**
    - advisor with rules engine
    - instant quote with add-on toggles
    - consent and Smart KYC (lookups, OCR, cross-checks)
    - order tracker
    - escalation to a named human
7. **Differentiating features:**
    - multilingual voice
    - WhatsApp updates
    - propensity-ranked admin dashboard
    - live metrics against baseline
8. **Human intervention strategy.** A human is brought in only when one of these happens:
    - a KYC check needs judgement (K06/K07-type cases);
    - a step is late against its expected date;
    - the advisor fails twice or the owner asks;
    - the deal is multi-site or a manufacturer.

    Field staff and partners take over these cases, plus High-propensity leads that didn’t finish. They get credit (referral codes, commission on app-completed sales), so the digital journey feeds them rather than competing with them.
9. **Revenue and pricing logic.** Catalogue prices only, no discounts. Micro is priced at the catalogue minimum because it’s very price-sensitive [Case]; other segments sit at the middle of the range. The quote shows one monthly total, the setup fee separately, and the contract length (the longest minimum contract in the bundle) [B].
10. **Success metrics.** Section 16.

How this differs from a conventional telecom CRM.

- **Who drives it.** A CRM records what salespeople do. This product *does the selling* for the standard case and hands people only the exceptions.
- **How eligibility works.** It enforces eligibility by rule, where 15% of bundles break the rules today [S].
- **Who sees status.** The customer sees the status, instead of the status living in an internal system.

Section 15 — Proposed Customer Journey



Step	Customer action	System action	Data needed	Output	Possible failure	Recovery	Human?
Discovery	Taps a link or scans a QR code	Opens the app in the referred context	Source, referral code	Session started	Link never opened	WhatsApp reminder (opt-in)	No

Step	Customer action	System action	Data needed	Output	Possible failure	Recovery	Human?
Registration	Enters phone, gets OTP, picks a language	Creates an account; records WhatsApp opt-in	Phone, language	Logged-in user	OTP not received	Resend; voice OTP	No
AI advisor	Answers 5–6 questions by text or voice	Builds the profile; asks for the GSTIN early	Business type, staff, sites, city, current setup, needs, GSTIN (optional)	Profile	Unclear answers; owner gives up	Rephrase; offer options; save and resume	If stuck twice
Recommendation	Reads the bundle and the reasons	Rules engine filters eligible products; AI explains	Profile, catalogue	Valid bundle	No eligible product (e.g. tier 2 + 1G request)	Explain why and offer the nearest valid option	No
Pricing	Toggles add-ons; sees the total	Recomputes the price; blocks invalid removals	Bundle, price rule	One monthly price	“Too expensive”	Show a smaller valid bundle; compare with current spend	For multi-site
Consent	Reads the notice and agrees	Stores consent (time, version, purpose)	—	Consent record	Declines	Explain what’s needed; allow save and exit	No
KYC	Enters the GSTIN; uploads 4 documents	Registry lookup pre-fills details; OCR reads documents	GSTIN, Udyam, 4 images	Extracted fields	Blurry or missing document	Instant re-upload prompt with a specific reason	No
Verification	Waits seconds	Runs every cross-check	Extracted fields, registries	Pass / fix / review / block	Mismatch; cancelled GSTIN	Specialist review; non-GST bundle	For mismatches
Order	Pays the first bill	Creates the order and schedules the install	Payment	Order ID	Payment fails	Retry; UPI alternative	No
Installation	Checks the tracker	Shows the expected date; sends proactive updates	Install days per product	Booked date	Late; site not feasible for fibre	Auto-escalate; AirFiber fallback	If late
Activation	Starts using the service	Marks it live; single bill; invites a referral	—	Active customer	Early dissatisfaction	Check-in message; referral offer after the first bill	No

Section 16 — Business Impact and Success Metrics

All targets below are proposals for a 6-month pilot. None has been achieved yet.

Metric	Definition	Baseline	Proposed target	How measured	Why it matters
Lead-to-activation	Activated ÷ leads created, by cohort	**7.4%** [P]	11%	Order and lead tables, weekly by segment and channel	The headline outcome
Average activation time	Lead created → live, in days	**17.6 days** [S]	8 days	Timestamps on order events	NPS falls as days rise (correlation −0.28) [S]
Customer acquisition cost	Sales and partner spend ÷ activations	**Rs 15,053** [S]	Rs 8,000	Spend ledger ÷ activations	Today's CAC is 2.2× the median monthly bill
Six-month churn	Share of activations lost within 6 months	**15.2%** [B]	11%	Billing status at 6 months	Lifetime value
Follow-up calls per customer	Customer-initiated status contacts	**3.0** [S]	under 0.5	Support contacts tagged "status"	Churn rises from 6% to 28% with chasing [S]
KYC completion rate	KYC complete ÷ quote accepted	**68.3%** (588/861) [F]	85%	Funnel events	All 273 KYC-stage losses are paperwork [F]
Document mismatch rate	Share of KYC submissions lost to mismatched details	**5.5%** of quote-accepted leads dropped for "details did not match" (47/861) [F]	under 2% lost; 100% of mismatches flagged at upload	KYC check results	Mismatches should be fixed in-session, not discovered later
Escalation resolution time	Escalation opened → resolved	**No baseline in the data.** The nearest proxy is 3.0 chase calls [F]	First response within 4 business hours; resolved within 24 hours	Escalation table timestamps	Exceptions are where human time is spent

How the main targets were derived [B]:

- **Conversion: 11%.** Professional offices already convert at 11.1% and referrals at 14.8%.
- **Days: 8.** About 9.0 of the 17.6 days pass before a quote is accepted, and 3.8 more for KYC. The core installs for micro and professional bundles take 5 days.
- **CAC: Rs 8,000.** Digital leads cost Rs 232 each; at 11% conversion that's about Rs 2,106 per self-serve activation. Field and partner deals cost Rs 22,965 today. A 50/50 mix, with assisted deals 40% cheaper because the app does discovery and KYC, gives about Rs 7,942.
- **Churn: 11%.** Customers who chased once churned at 6% and twice at 14%.

Assumptions to validate: the 50/50 mix, the 40% saving on assisted deals, and the KYC completion target are assumptions. The pilot should measure them.

Section 17 — Technical Implications

This section turns findings into requirements. It deliberately avoids locking in technology choices.

Area	What the findings require
Frontend	Mobile-first, works on low-end phones and slow networks. All text in language files (7 languages cover 94.2% of leads [F]). Large tap targets. Status as the post-purchase home screen
Backend services	Separate services or modules for: advisor, rules and pricing, KYC, orders and tracker, escalation, notifications, admin. Each step writes an event, so the funnel metrics are computable
AI advisor	Language model for the conversation only: asking, summarising, explaining. It never decides eligibility or price. It asks for the GSTIN early. It stays inside the catalogue
Rules engine	Deterministic code enforcing tier, minimum employees, minimum sites, active GSTIN and dependencies, returning a reason for every rejection. Already prototyped as `validate_bundles.py`, with 31 passing checks [B]. Must pass the judges' 15 live scenarios [Case]
Product recommendation	Start from the segment's validated starter bundle, then add the products the owner asked for if eligible. Per-site and per-user quantities scale with the profile
Pricing	One documented rule (micro at the minimum, others at the midpoint), setup fee separate, contract = longest minimum contract. Recompute on every toggle
OCR	Watermark and colour cleanup, deskew, blur detection, GSTIN checksum repair against the registry, word-level name matching. Already prototyped: 6 of 12 flagged, 6 verified [K]. Low-confidence reads go to "confirm" or review, never silent acceptance
Registry APIs	Mock GSTIN and Udyam lookups backed by the CSVs. Never match on name alone; use the number, or name plus city/state
Database	Leads, users, conversations, quotes, orders, order events, KYC cases and checks, consents, escalations, agents, notifications. Personal identifiers masked in UI and logs; documents kept for a stated period
Propensity model	Logistic regression served from a saved model, returning score, bucket and top 3 reasons [M]. Retrain on app data
Notifications	Opt-in WhatsApp/SMS for quote, KYC result, install booked, delay, live, and incomplete-KYC reminders. A mock outbox for the demo
Voice	Speech-to-text and text-to-speech behind one interface, so providers can be swapped. Short spoken responses
Admin dashboard	Unclosed leads with propensity and reasons; escalation queue; the 5 metrics against baseline, live

Section 18 — Risks and Limitations

Risk	What could happen	Mitigation
Data quality	Hidden errors skew decisions	40 consistency checks all passed [P]. Every number is reproducible from scripts
Synthetic data	Patterns are planted, not observed in the market. Some look odd: promoters churn more (19%) than detractors (13%) [F]; referral customers churn most (24.6%) [F]	Treat findings as directional. Validate in a pilot. Don't over-fit the product to quirks
Small samples	37 activated manufacturers; 12 KYC applicants	Report counts alongside percentages; avoid strong claims on small groups
OCR errors	Raw OCR read only 1 of 11 GSTINs exactly [K]	Checksum and registry repair. "Confirm the number" step. Low-confidence reads go to review. Ask the user to type the GSTIN and use OCR as a cross-check
AI hallucination	The advisor invents a product, price or promise	The AI only explains outputs from the rules engine and pricing. Product IDs and prices come from code, never from the model
Wrong recommendations	An ineligible or oversized bundle	Rules engine with 31 automated checks [B]. The judges' 15 scenarios become the regression test
Privacy and consent	KYC data collected without valid consent; identifiers leak	Consent before any upload. Collect only what each step needs. Mask GSTIN, Udyam and ID in UI and logs. Stated retention and auto-delete. Role-based access. Audit log
External APIs	Registry, OCR, voice or WhatsApp providers fail	Mocks for the demo. Provider interfaces with fallbacks (manual entry, text chat)
WhatsApp limits	Opt-in rules, template approval, delivery failures	Opt-in checkbox; in-app tracker as the source of truth; SMS fallback
Agent availability	Escalations pile up	Escalate only on defined triggers. Least-loaded, language-matched assignment. Response-time metric on the dashboard
Model limits	Scores reflect the old process and could steer effort unfairly	Use only to order calls, never to deny service. Show reasons. Retrain on app data
Unknown answer key	Judges' KYC key or advisor tests include cases we didn't model	Compare against the key once released. Keep checks explainable and easy to extend

Section 19 — Final Team Decision Framework

A. What we know from the data

- Conversion is 7.4%, the quote step passes 43.6%, and activation takes 17.6 days with 3.0 chase calls [P][F][S].
- 41.2% of drops are about reach and speed; 32.4% are directly addressable and 36.8% partly [S].
- A GSTIN raises activation from 2.0% to 9.6% [S].
- 15% of today's bundles break the catalogue rules [S].
- Chase calls track churn: 6% at 1 call, 28% at 5+ [S].
- Professional offices have the lowest CAC (Rs 10,695); multi-site the highest value (Rs 28,080); micro the highest churn (22.7%) [F].
- Referral is the most efficient channel; field sales the most expensive [S].
- 6 of 12 KYC applicants have planted problems, all caught by our checker [K].

B. What we believe but still need to validate

- A self-serve advisor will raise quote acceptance.
- The tracker will cut chasing, and with it churn.
- Assisted deals will cost about 40% less when the app does discovery and KYC.
- Regional-language voice will narrow the conversion gap for Tamil, Punjabi and Kannada speakers.
- Existing Accel mobile customers (37.8% of leads, 9.2% conversion) can be upgraded through a pre-filled journey.

C. What the team must decide

1. Confirm the hybrid direction (B) and professional offices as the launch segment.
2. How many languages ship in the demo, and is voice in the core demo or a stretch?
3. In the KYC step, does the user type the GSTIN (OCR cross-checks it), or does OCR read it (the user confirms)?
4. Exact escalation triggers and response-time targets.
5. Which of WhatsApp reminders, the admin dashboard and propensity make the 3-minute demo.

D. What we should build first

The rules engine and pricing, then the advisor on top, then consent and Smart KYC, then the tracker with escalation. Each piece is testable on its own, and together they form the demo's main flow.

E. What we should deliberately avoid

- Letting the language model choose products or prices.
- Dynamic or personalised discounts.
- Auto-rejecting KYC on anything except a cancelled or suspended GSTIN.
- Name-only registry lookups.
- Using the propensity score to refuse service.
- Copying the real Airtel brand inside the product UI (the case uses "Accel").

F. What can be postponed

- Real payment gateway (mock it).
- Real WhatsApp Business API (sandbox or mock outbox).
- Referral programme mechanics.
- Existing-customer upgrade journey.
- E-sign and DigiLocker integration.
- More languages.

Prioritisation matrix

Must have	Should have	Could have	Won't have (this hackathon)
Rules engine (eligibility + dependencies)	Voice in Hindi + English	Existing-customer upgrade flow	Dynamic pricing

Must have	Should have	Could have	Won’t have (this hackathon)
Advisor chat with plain-language “why”	WhatsApp updates and KYC reminders (mock OK)	Referral links and partner codes	Real payment gateway
Instant quote with add-on toggles	Admin dashboard: unclosed leads + propensity	Regional voice beyond Hindi	Real DigiLocker / e-sign
Consent screen before KYC	Live metrics against baseline	Agent mobile view	Native mobile apps
GSTIN/Udyam mock lookups + OCR checks	5 regional languages in text	Partner commission tracking	Full CRM
Order tracker with expected dates			
Escalation to a named agent			

Five questions for the team discussion

1. If a judge’s advisor scenario conflicts with our starter bundle, do we trust the rules engine’s eligible set or the segment’s typical bundle? (Our proposal: rules first, typical bundle as the default within them.)
2. Should the GSTIN lookup come *before* the recommendation, since it decides 10 of the 25 products? Or after, to keep the first experience light?
3. For micro, do we lead with the Rs 1,597 core and upsell, or show the Rs 2,794 bundle with add-ons pre-selected?
4. What exactly does a field agent see and get paid for when they pick up an escalated or High-propensity lead?
5. Which single moment in the 3-minute demo proves “humans handle exceptions”? The K06 GSTIN mismatch going to review, or a late install auto-escalating?

Section 20 — Final Recommendation

Build the hybrid Udyam Digital journey: an AI advisor backed by a rules engine, instant plain-language pricing, consent-first Smart KYC, and a live tracker, with automatic escalation to a named human for exceptions. Launch it for professional offices, and run multi-site as an assisted track.

- **Why this solution.**
  - The losses before the quote (65% of all losses) are about waiting and visits.
  - The losses after it are exceptions: KYC flags and installs [F][K].
  - Only the hybrid addresses both.
- **Why this segment.**
  - Professional offices convert at 11.1% and have the lowest CAC (Rs 10,695).
  - 86% have a GSTIN, and their core installs in 5 days [F][B].
  - That makes them the best place to prove self-serve before scaling to micro.
- **Why these features.** Each one answers a measured problem:
  - advisor → visits and reach (827 + 805 drops)
  - instant quote → the 43.6% step
  - rules engine → 15% mis-selling
  - Smart KYC → 273 paperwork drops
  - tracker → chasing and churn (6% → 28%)
  - escalation → cases software can’t settle (K06, K07)

  Sources: [S][F][K].
- **What makes it different.** Rules decide, AI explains, and people handle exceptions. The customer always sees their own status.
- **Target outcome (pilot targets, not results).**

Metric	Today	Target
Conversion	7.4%	11%
Days to activate	17.6	8
CAC	Rs 15,053	Rs 8,000
6-month churn	15.2%	11%
Chase calls per customer	3.0	under 0.5

  Source: [B].

One-page executive decision summary

Problem.

- Small businesses buy Accel's services through visits, slow quotes, paper KYC and a 17.6-day wait.
- Only 7.4% of 5,000 leads became customers, at Rs 15,053 each [P][S].

Where it breaks.

Point in the journey	What the data shows
Quote	Only 43.6% accept it (32% for micro) [F][S]
Reach and speed	41.2% of all drops [S]
Paperwork	All 273 KYC-stage drops [F]
Status chasing	Churn 6% at 1 call vs 28% at 5+ [S]
Catalogue rules	15% of sold bundles break them [S]

Decision. Build **Option B, the hybrid** (weighted score 4.45 vs 3.45 for AI-only and 2.95 for an agent tool; team judgement, Section 13).

Launch segment. Professional offices: 11.1% conversion, Rs 10,695 CAC, 86% with a GSTIN, 5-day install. Multi-site runs as an assisted track at a Rs 28,080 median bill. Micro follows in phase 2 [F][B].

Must-build features:

1. Rules engine. 22 bundle checks and 9 must-refuse checks already pass [B].
2. Advisor that explains the bundle in plain words.
3. Instant quote with add-on toggles.
4. Consent screen, then Smart KYC. 47 documents already processed; 6 of 12 applicants flagged, 6 verified [K].
5. Tracker with expected dates.
6. Escalation to a named human.

Differentiators. Multilingual voice; WhatsApp updates; propensity-ranked admin dashboard. The top 20% of leads hold 43% of activations [M].

Pilot targets (not results). Conversion 11%, 8 days, Rs 8,000 CAC, 11% churn, under 0.5 chase calls, 85% KYC completion.

Biggest risks.

- AI inventing products or prices: rules decide, AI explains.
- OCR misreads: checksum repair, confirm step, review.
- Agent capacity: escalate on defined triggers only.
- Synthetic data: validate in the pilot.

Next step. Approve this direction, then start Phase 2: architecture, database schema, API contracts and screen-by-screen design, all built around this journey.

Reproduce every number: `python analysis/01_profile.py`, `02_signals.py`, `03_bundles_metrics.py`, `03_propensity.py`, `05_report_facts.py`, `06_charts.py`, `validate_bundles.py`, and `python scripts/kyc_check.py .` from the repository root.

Part B — Udyam Digital System Design (DESIGN.md)

This is the build specification for Phases 1–5. It is written before any production code. It follows from the Phase 1 report: every design choice below points back to a finding there, tagged the same way ([S], [F], [B], [M], [K]).

Design principles

1. **Rules decide, AI explains.** The language model runs the conversation and writes the “why”. Eligibility, quantities and prices come only from deterministic code. Today 15% of sold bundles break catalogue rules [S], and the judges test eligibility live [Case].
2. **AI handles acquisition; humans handle exceptions.** Escalation is a first-class part of the system, not an afterthought. Half the KYC sample needed a fix or a review [K].
3. **Consent before KYC, minimum data, masked identifiers.** Required by the case and the DPDP Act 2023 [Case].
4. **The customer always sees their own status.** Chase calls track churn: 6% at 1 call, 28% at 5+ [S].
5. **Everything writes an event.** This keeps the five pilot metrics computable from the database [B].

What already exists from Phase 1 and gets reused

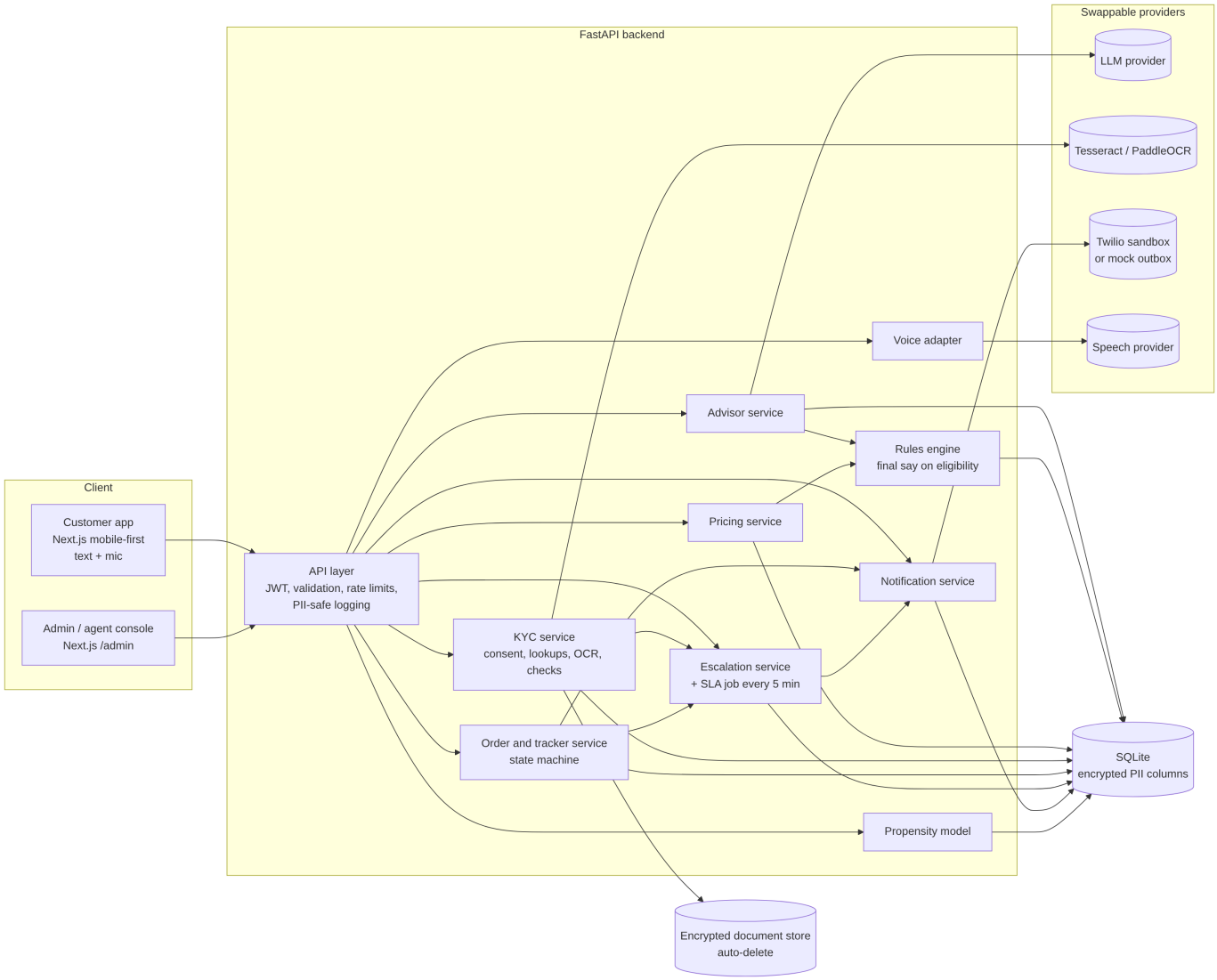
Asset	Becomes
`analysis/validate_bundles.py` (`rule_errors`, `price`, `scale`)	`backend/app/rules/engine.py` and `backend/app/pricing/service.py`
`analysis/bundles.json` (4 validated starter bundles)	`backend/app/rules/bundles.json`, the advisor’s default bundle per segment
`scripts/kyc_check.py` (watermark removal, deskew, blur check, GSTIN checksum repair, word-level name match, all cross-checks)	`backend/app/kyc/ocr.py` and `backend/app/kyc/checks.py`
`backend/app/propensity.py` + `backend/models/propensity.pkl`	Used as-is by the admin leads endpoints
31 rules checks, 12 KYC verdicts	The first automated test suites

D1. Architecture

D1.1 Components

Component	Responsibility	Talks to
Frontend (Next.js, Tailwind)	Mobile-first customer app; admin/agent console under `/admin`; i18n for 7 languages (94.2% of leads [F]); browser speech APIs for voice	API over HTTPS (JSON + multipart)
API gateway (FastAPI)	Auth (JWT), routing, validation (Pydantic), rate limits, error format, PII-safe logging	All services (in-process modules for the hackathon)
Advisor service	Runs the conversation; builds the business profile; calls the rules engine; asks the LLM to explain only what the rules returned; counts failed turns	LLM provider interface, rules engine, DB
Rules engine	`eligible_products(profile)`, `validate_bundle(profile, items)`, `default_bundle(segment, profile)`. Pure functions, no I/O	Products table, `bundles.json`
Pricing service	Unit price rule, totals, setup fee, contract length, install days; recalculates on every toggle	Rules engine, products
KYC service	Consent records; GSTIN/Udyam mock lookups; upload storage; OCR pipeline; cross-checks; verdict	Registries, OCR engine, file store, escalation service
Order & tracker service	Order state machine, expected dates, tracker view, install simulator	Products (install days), notifications, escalation service
Escalation service + job	Creates escalations from triggers; least-loaded, language-matched assignment; SLA timers; periodic scan for overdue states	Orders, KYC, conversations, agents, notifications
Notification service	Templates per language; WhatsApp/SMS via Twilio sandbox or a mock outbox; opt-in enforced	Users, notifications table, provider
Voice service	Browser Web Speech API first; server adapter (Bhashini/Sarvam/other) behind `/voice/*` as fallback	Advisor
Propensity model	`predict(lead)` returns score, bucket and top 3 reasons [M]	Leads
Admin dashboard	Leads table with propensity, lead detail with call and notes, escalation queue, KYC review, metrics strip, notification outbox, order simulator	Admin endpoints
Data stores	SQLite (SQLAlchemy); encrypted file store for documents; no external database for the demo	—

D1.2 Architecture diagram



D1.3 Main data flows

1. Discovery to quote

```
Customer message -> POST /advisor/message
  -> Advisor updates profile_json and asks the next question (max 6)
  -> when the profile is complete: rules.eligible_products(profile)
       + rules.default_bundle(segment) + products the owner asked for, if eligible
  -> rules.validate_bundle()  (must return no errors)
  -> LLM explains the validated items in the user's language
       (the prompt contains only validated items; the output is checked
        so it names no product_id outside the list)
  -> recommendation saved -> POST /quotes -> pricing.price(items)
```

2. Quote to order

```
Toggle add-on -> PATCH /quotes/{id}/items/{pid}
  -> validate_bundle; if removing an item breaks a dependency, it is locked with a reason
  -> totals recalculated -> POST /quotes/{id}/accept -> POST /orders (QUOTE_ACCEPTED)
```

3. KYC

```
POST /consent (purpose=kyc_verification) -> order CONSENT_GIVEN, kyc_case opened
-> GET /kyc/gstin/{gstin} (registry pre-fill; a cancelled/suspended GSTIN removes
   GST-only items, and the quote is re-priced)
-> POST /kyc/upload x4 (OCR runs on upload: blur, deskew, checksum repair)
-> POST /kyc/{case}/submit -> all cross-checks -> verdict
     Verified          -> KYC_VERIFIED
     Re-upload needed  -> KYC_FLAGGED, specific fix-it messages
     Human review      -> KYC_FLAGGED -> HUMAN_REVIEW + escalation
     Blocked           -> KYC_FLAGGED, offer non-GST bundle
```

4. Fulfilment

```
KYC_VERIFIED -> INSTALL_BOOKED (expected date = verified_at + max install days)
  -> INSTALLED -> LIVE
Every transition: order_event + notification (if opted in) + tracker update
SLA job: an overdue state creates an escalation, assigns an agent, notifies the user
```

5. Admin

```
GET /admin/leads -> leads + propensity bucket and reasons
  -> POST /admin/leads/{id}/call (call_log)
  -> POST /admin/leads/{id}/notes
GET /escalations -> agent picks up -> PATCH resolve -> order/KYC continues
```

D1.4 Advisor guardrails (from the judges' 15 hidden scenarios)

- The LLM receives a tool schema and can only call `ask_question`, `set_profile_field`, `request_recommendation` or `handoff_to_human`.
- It never receives prices to invent. It receives the validated items and their prices from the pricing service.
- Every LLM reply is post-checked:
  - any product ID or price not in the validated set → reply discarded and regenerated;
  - after 2 failures → a template reply is used.
- A **failed turn** is any of the following. Two failed turns in one conversation trigger `advisor_failed` escalation:
  - the LLM's output fails validation twice;
  - the user repeats an answer the advisor couldn't parse;
  - the user says "talk to a person" (or the equivalent in their language).
- Ineligible requests always get a reason and what would unlock them (e.g. "WhatsApp Business needs an active GSTIN"). The 9 must-refuse cases from [B] become fixed test cases.
- The GSTIN question is asked early (question 2 or 3), because it gates 10 of 25 products and lifts activation from 2.0% to 9.6% [S][F].

D2. Database schema (SQLite)

Conventions

- **Keys and timestamps.** Primary keys are `TEXT` UUIDv7 unless noted. Timestamps are ISO-8601 UTC `TEXT`. Money is `INTEGER` rupees. JSON is stored as `TEXT` (validated by Pydantic).
- **Personal data marking.** `PII` marks personal data. `[enc]` = encrypted with Fernet (AES-128-CBC + HMAC) using `APP_ENCRYPTION_KEY`. `[hash]` = HMAC-SHA256 with `PII_HASH_KEY`, used only for lookups. `[mask]` = a display-only masked copy, safe for UI and logs.
- **Masking formats:**

Identifier	Masked form
Phone	`••••••4321`
GSTIN	`27•••••••••1Z5` (first 2 + last 3)
Udyam number	`UDYAM-MH-00-•••4567`
ID numbers	last 4
Personal names in admin lists	first name + initial

D2.1 Identity and access

users

Column	Type	Key / constraint	PII
id	TEXT	PK	
phone_enc	BLOB	NOT NULL	PII [enc]
phone_hash	TEXT	UNIQUE, NOT NULL	PII [hash]
phone_masked	TEXT		[mask]
display_name_enc	BLOB	NULL	PII [enc]
role	TEXT	CHECK in (customer, agent, admin)	
language	TEXT	default ‘hi’	
whatsapp_opt_in	INTEGER	0/1, default 0	
created_at, last_login_at	TEXT		

otp_requests (mock OTP)

Column	Type	Key	PII
id	TEXT	PK	
phone_hash	TEXT	INDEX	PII [hash]
code_hash	TEXT		(never the plain code)
expires_at	TEXT		
attempts	INTEGER	max 5	
consumed_at, created_at	TEXT		

agents

Column	Type	Key	PII
id	TEXT	PK	
user_id	TEXT	FK → users.id, UNIQUE	
name	TEXT		staff name (not customer PII)
languages	TEXT (JSON)	e.g. [“hi”,“en”,“ta”]	
skills	TEXT (JSON)	[“kyc”,“sales”,“install”,“account_manager”]	
max_open	INTEGER	default 8	

Column	Type	Key	PII
active	INTEGER	0/1	
created_at	TEXT		

audit_log

Column	Type	Key	PII
id	INTEGER	PK autoincrement	
actor_user_id	TEXT	FK → users.id	
action	TEXT	view_document, view_pii, resolve_escalation, call_lead, …	
entity, entity_id	TEXT		
created_at	TEXT		

D2.2 Business, leads and reference data

businesses

Column	Type	Key	PII
id	TEXT	PK	
user_id	TEXT	FK → users.id, INDEX	
name	TEXT		business name (not personal)
business_type, segment	TEXT	segment CHECK in 4 values	
city, state	TEXT		
city_tier	INTEGER	CHECK in (1,2)	
employees, sites	INTEGER	≥ 1	
turnover_band, current_internet, current_provider	TEXT		
monthly_spend_inr	INTEGER		
has_gstin	INTEGER	0/1	
gstin_enc	BLOB		PII [enc] (proprietor GSTINs embed a personal PAN)
gstin_masked	TEXT		[mask]
gst_status	TEXT	Active / Cancelled / Suspended / None	
has_udyam	INTEGER		
udyam_enc / udyam_masked	BLOB / TEXT		PII [enc] / [mask]
products_of_interest	TEXT (JSON)		
lead_source, referral_code	TEXT		
created_at, updated_at	TEXT		

leads (5,000 rows imported from `smb_leads.csv`, plus every new app lead)

Column	Type	Key	PII
lead_id	TEXT	PK (L00001… for imports, L-APP-… for new)	
origin	TEXT	import / app	
business_id	TEXT	FK → businesses.id, NULL for imports	
all CSV columns	as in the CSV (dates TEXT, money INTEGER)		contact_name is PII [mask] in UI; gstin/udyam_number [mask] in UI

Column	Type	Key	PII
propensity_score	REAL		
propensity_bucket	TEXT	High / Medium / Low, INDEX	
propensity_reasons	TEXT (JSON)	top 3	
advisor_outcome	TEXT	not_started / abandoned / escalated / quoted / ordered	
assigned_agent_id	TEXT	FK → agents.id	
last_contacted_at	TEXT		
lead_status	TEXT	open / contacted / converted / closed	

products (25 rows from `product_catalogue.csv`)

Column	Type	Key
product_id	TEXT	PK (P01…P25)
category, name, description, unit	TEXT	
price_min_inr, price_max_inr, setup_fee_inr	INTEGER	
min_contract_months, min_employees, min_sites, install_days	INTEGER	
city_tiers	TEXT (JSON)	e.g. [1,2]
requires_gstin	INTEGER	0/1
requires_one_of	TEXT (JSON)	e.g. [“P01”,“P02”,“P03”]
target_segments	TEXT (JSON)	
quantity_basis	TEXT	per_site / per_user / fixed (from `bundles.json`)

gstin_registry (3,534 rows)

Column	Type	Key
gstin	TEXT	PK
legal_name, trade_name	TEXT	INDEX on lower(trade_name)
constitution, principal_place_of_business, state	TEXT	
registration_date	TEXT	
status	TEXT	Active / Cancelled / Suspended

Synthetic, so stored in plain text. A real deployment would call the GSTN API rather than store it.

udyam_registry (2,723 rows)

Column	Type	Key
udyam_number	TEXT	PK
enterprise_name	TEXT	INDEX (lower(enterprise_name), lower(city))
enterprise_type, major_activity, nic_description, state, city	TEXT	
date_of_registration	TEXT	

Composite index on name + city, because name-only lookups pick the wrong business [K].

D2.3 Advisor

conversations

Column	Type	Key	PII
id	TEXT	PK	
user_id	TEXT	FK → users.id	
business_id	TEXT	FK → businesses.id, NULL until the profile is saved	
lead_id	TEXT	FK → leads.lead_id	
language, channel	TEXT	channel: text / voice	
status	TEXT	active / recommended / escalated / abandoned	
profile_json	TEXT (JSON)		GSTIN stored masked here
questions_asked	INTEGER		
failed_turns	INTEGER	default 0; 2 triggers escalation	
created_at, updated_at	TEXT		

messages

Column	Type	Key	PII
id	TEXT	PK	
conversation_id	TEXT	FK, INDEX	
role	TEXT	user / assistant / system / tool	
content	TEXT		PII scrubbed before saving (regex for GSTIN, Udyam, phone, PAN replaced with masked forms)
modality	TEXT	text / voice	
audio_ref	TEXT	NULL	not stored by default
validation_status	TEXT	ok / regenerated / template	
created_at	TEXT		

recommendations

Column	Type	Key
id	TEXT	PK
conversation_id	TEXT	FK
business_id	TEXT	FK
bundle_key	TEXT	micro / prof / multi / mfg / custom
items	TEXT (JSON)	[{product_id, qty, role: core/addon}]
blocked	TEXT (JSON)	[{product_id, reasons:[...]}] (asked for but ineligible)
explanation	TEXT	LLM text, in the user's language
rules_version	TEXT	git SHA of `rules/engine.py` + `bundles.json`
created_at	TEXT	

D2.4 Quotes and orders

quotes

Column	Type	Key
id	TEXT	PK
business_id	TEXT	FK
recommendation_id	TEXT	FK

Column	Type	Key
status	TEXT	draft / accepted / superseded / expired
monthly_total_inr, setup_total_inr	INTEGER	
contract_months, max_install_days	INTEGER	
pricing_rule	TEXT	“micro_min_v1” (micro: catalogue minimum) / “midpoint_v1” (others: rounded midpoint of min–max); see docs/PRICING.md
current_spend_inr	INTEGER	for the “vs what you pay now” line
valid_until	TEXT	created + 7 days
accepted_at, created_at	TEXT	

quote_items

Column	Type	Key
id	TEXT	PK
quote_id	TEXT	FK, INDEX; UNIQUE(quote_id, product_id)
product_id	TEXT	FK → products
qty	INTEGER	≥ 1
role	TEXT	core / addon
selected	INTEGER	0/1 (add-on toggle)
unit_price_inr, line_monthly_inr, line_setup_inr	INTEGER	
locked	INTEGER	0/1: can’t be removed because another selected item depends on it
lock_reason	TEXT	e.g. “Static IP needs Biz Fibre”

orders

Column	Type	Key
id	TEXT	PK (human-friendly `UD-2607-0042` shown in UI)
quote_id	TEXT	FK, UNIQUE
business_id, user_id	TEXT	FK
state	TEXT	CHECK in the state list (D4), INDEX
state_entered_at	TEXT	
state_due_at	TEXT	SLA deadline for the current state, INDEX (scanned by the job)
expected_install_date, expected_live_date	TEXT	
install_slot	TEXT	e.g. “2026-07-21 10:00-12:00”
assigned_agent_id	TEXT	FK → agents (when escalated)
payment_status	TEXT	pending / paid / failed (mock gateway)
created_at, updated_at	TEXT	

order_events

Column	Type	Key
id	INTEGER	PK autoincrement
order_id	TEXT	FK, INDEX
from_state, to_state	TEXT	
actor	TEXT	system / user / agent:{id} / simulator
note	TEXT	

Column	Type	Key
created_at	TEXT	

D2.5 Consent and KYC

consents

Column	Type	Key	PII
id	TEXT	PK	
user_id	TEXT	FK	
business_id	TEXT	FK	
order_id	TEXT	FK, NULL for WhatsApp consent	
purpose	TEXT	kyc_verification / whatsapp_updates	
notice_version	TEXT	e.g. “kyc-v1.0”	
language	TEXT		
granted	INTEGER	0/1	
granted_at, withdrawn_at	TEXT		
client_hash	TEXT		[hash] of IP + user agent (proof, not tracking)

kyc_cases

Column	Type	Key	PII
id	TEXT	PK	
order_id	TEXT	FK, UNIQUE	
business_id	TEXT	FK	
consent_id	TEXT	FK → consents.id, NOT NULL (no case without consent)	
status	TEXT	open / submitted / verified / flagged / human_review / rejected	
verdict	TEXT	Verified / Re-upload needed / Human review / Blocked	
gstin_enc, gstin_masked	BLOB / TEXT		PII [enc] / [mask]
udyam_enc, udyam_masked	BLOB / TEXT		PII [enc] / [mask]
registry_snapshot	TEXT (JSON)	trade name, status, state at check time	
retention_until	TEXT	verified_at + 90 days (or closed + 30 days if abandoned)	
created_at, submitted_at, verified_at	TEXT		

kyc_documents

Column	Type	Key	PII
id	TEXT	PK	
case_id	TEXT	FK, INDEX; UNIQUE(case_id, doc_type, version)	
doc_type	TEXT	gst_certificate / udyam_certificate / signatory_id / address_proof	
version	INTEGER	re-uploads increment it	

Column	Type	Key	PII
storage_path_enc	BLOB		PII [enc]; file itself encrypted at rest
sha256	TEXT	duplicate/tamper check	
mime, size_bytes	TEXT / INTEGER		
ocr_text_enc	BLOB		PII [enc]
ocr_confidence, sharpness, skew_deg	REAL		
extracted	TEXT (JSON)		fields stored masked (names kept for matching, IDs masked)
status	TEXT	uploaded / processed / superseded / deleted	
uploaded_at, deleted_at	TEXT		

kyc_checks

Column	Type	Key
id	INTEGER	PK autoincrement
case_id	TEXT	FK, INDEX
document_id	TEXT	FK, NULL for registry-only checks
check_code	TEXT	e.g. GSTIN_CHECKSUM, GSTIN_ACTIVE, NAME_TRADE_MATCH, STATE_CODE, UDYAM_FOUND, UDYAM_NAME_CITY, BILL_AGE_90D, BILL_NAME, BILL_ADDRESS, SIGNATORY_BUSINESS, ID_EXPIRY, SIGNATORY_IS_PROPRIETOR, DOC_PRESENT, DOC_READABLE
result	TEXT	pass / fail
severity	TEXT	blocking / unreadable / missing_document / stale / mismatch / confirm / needs_review / NULL
message_customer	TEXT	in the user’s language, with a specific fix
detail_internal	TEXT	masked
created_at	TEXT	

D2.6 Operations

escalations

Column	Type	Key
id	TEXT	PK
trigger	TEXT	sla_breach / kyc_flag / advisor_failed / user_request / assisted_deal
priority	TEXT	P1 / P2 / P3
order_id, kyc_case_id, conversation_id, lead_id	TEXT	FKs, at least one NOT NULL (CHECK)
reason	TEXT	human-readable, masked
status	TEXT	open / assigned / in_progress / resolved / cancelled, INDEX
assigned_agent_id	TEXT	FK → agents
required_skill, language	TEXT	used for matching
sla_due_at	TEXT	first-response deadline
created_at, assigned_at, first_response_at, resolved_at	TEXT	

Column	Type	Key
resolution	TEXT	approved / rejected / customer_contacted / rebooked / other
resolution_note	TEXT	

UNIQUE(trigger, order_id) WHERE status not in (resolved, cancelled), to avoid duplicates.

notifications

Column	Type	Key	PII
id	TEXT	PK	
user_id	TEXT	FK	
order_id	TEXT	FK, NULL	
channel	TEXT	whatsapp / sms / in_app	
template	TEXT	quote_ready, kyc_verified, kyc_fix_needed, kyc_review, install_booked, delay_escalated, live, kyc_reminder_24h, agent_assigned	
language	TEXT		
payload	TEXT (JSON)		masked values only
status	TEXT	queued / sent / delivered / failed / skipped_no_optin	
provider_ref, error	TEXT		
created_at, sent_at	TEXT		

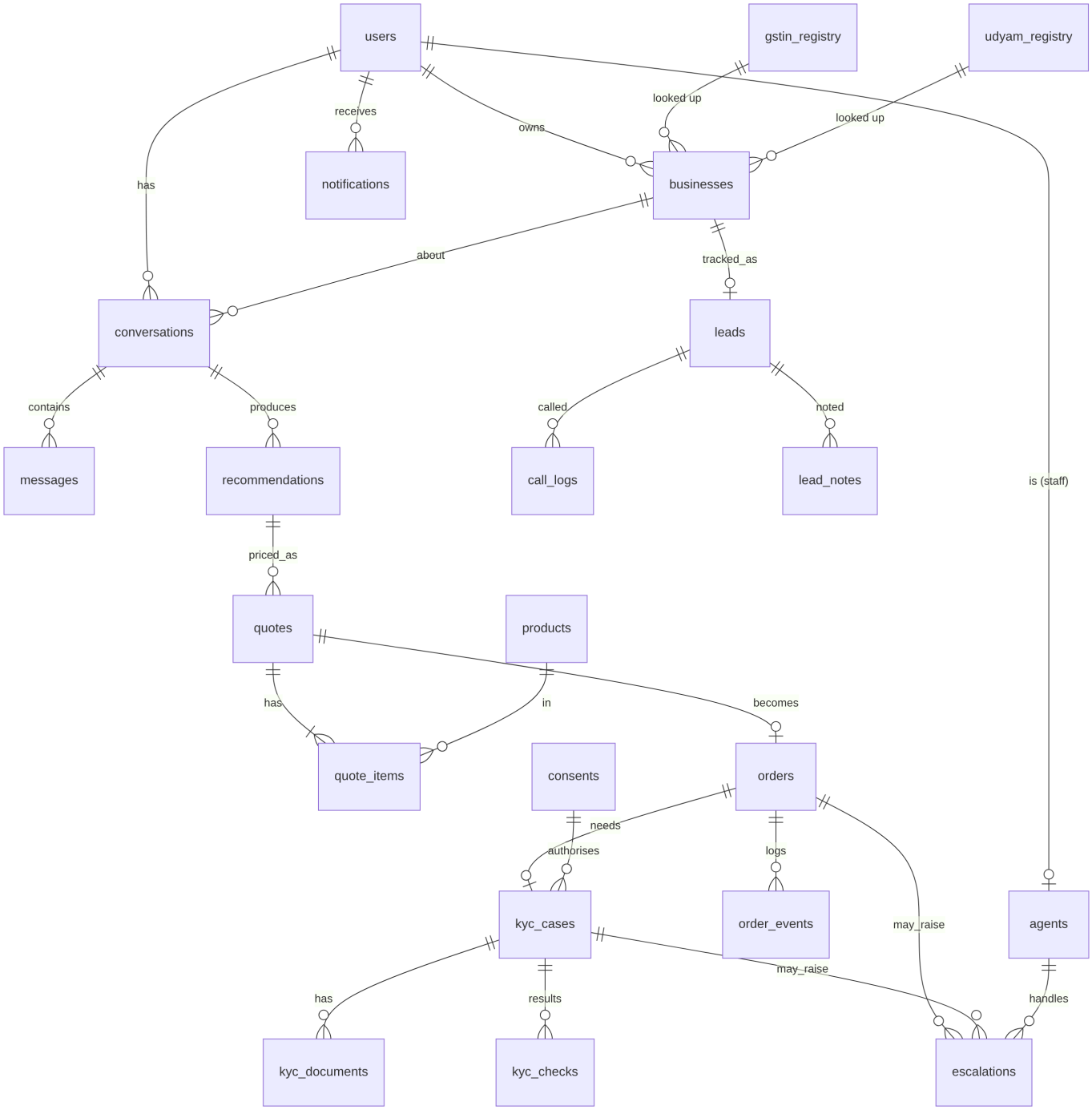
call_logs

Column	Type	Key
id	TEXT	PK
lead_id	TEXT	FK, INDEX
agent_id	TEXT	FK
provider	TEXT	twilio / mock
outcome	TEXT	connected / no_answer / callback_requested / interested / converted / not_interested
duration_s	INTEGER	
provider_ref	TEXT	
created_at	TEXT	

lead_notes

Column	Type	Key
id	TEXT	PK
lead_id	TEXT	FK, INDEX
agent_id	TEXT	FK
body	TEXT	free text, PII-scrubbed on save
created_at	TEXT	

D2.7 Relationships



D2.8 Data protection rules

Rule	Implementation
Consent before KYC	`kyc_cases.consent_id NOT NULL`; `POST /kyc/upload` returns `403 CONSENT_REQUIRED` without an active consent
Data minimisation	Advisor stores no IDs except an optional GSTIN (masked in profile). KYC stores only the 4 required documents. Voice audio not stored
Encryption at rest	PII columns `[enc]`; document files encrypted with a per-file key wrapped by `APP_ENCRYPTION_KEY`
Masking in UI and logs	API serialisers return only `*_masked` fields. A logging filter redacts GSTIN, Udyam, PAN and phone patterns
Retention	Nightly job deletes document files and `ocr_text_enc` after `retention_until` (90 days after verification, 30 after abandonment) and sets `kyc_documents.status = deleted`

Rule	Implementation
Access	Customer sees only their own records. Agents see assigned cases. Unmasked document view needs the agent or admin role and writes `audit_log`
Withdrawal	`DELETE /consent/{id}` stops processing, cancels any open KYC case and schedules deletion

D3. API contracts

Base and auth:

- Base URL: `/api/v1`.
- Send `Authorization: Bearer <jwt>` on every request except `auth/*`, `GET /products` and `GET /health`.
- JWT claims: `sub` (user id), `role`, `lang`, and a 24 h expiry.

Error format, used everywhere:

```
        { "error": { "code": "GSTIN_INVALID", "message": "This GST number isn't valid. Please check it.", "details": {
"field": "gstin" } } }
```

`message` is localised to the user's language.

Common errors:

Status	Code
400	`VALIDATION_ERROR`
401	`UNAUTHENTICATED`
403	`FORBIDDEN`
404	`NOT_FOUND`
409	`INVALID_STATE`
429	`RATE_LIMITED`
503	`PROVIDER_UNAVAILABLE` (LLM, OCR or messaging down; the response carries a fallback)

Under each endpoint, "Tables" lists what it reads (R) and writes (W).

D3.1 Auth

`POST /auth/otp/request`

Request `{ "phone": "+919876543210" }`

Response `200 { "request_id": "otp_01J…", "expires_in": 300, "dev_code": "123456" }`. `dev_code` is returned only when `OTP_MODE=mock`.

Errors: `400 PHONE_INVALID`, `429 RATE_LIMITED` (3 per 10 min per phone).

Tables: W `otp_requests`.

`POST /auth/otp/verify`

Request `{ "request_id": "otp_01J…", "code": "123456", "language": "hi", "whatsapp_opt_in": true }`

Response `200 { "token": "<jwt>", "user": { "id": "usr_…", "role": "customer", "language": "hi", "phone_masked": "••••••3210", "whatsapp_opt_in": true } }`

Errors: `400 OTP_INVALID`, `410 OTP_EXPIRED`, `429 OTP_TOO_MANY_ATTEMPTS`.

Tables: R/W `otp_requests`; W `users` (created on first login); W `consents` (purpose `whatsapp_updates` when opted in).

`GET /me` and `PATCH /me`

PATCH request `{ "language": "ta", "whatsapp_opt_in": false }`

Response `200 { "user": {…}, "business": { "id": "biz_…", "name": "…", "segment": "Professional offices", "gstin_masked": "27•••••••••1Z5" } | null }`

Tables: R/W `users`, `consents`; R `businesses`.

`POST /auth/logout`

Response `204`. The token is added to a short-lived deny list (in memory).

Tables: none (no database write).

D3.2 Advisor and voice

`POST /advisor/conversations`

Starts or resumes a session.

Request `{ "language": "hi", "channel": "text", "lead_source": "WhatsApp campaign", "referral_code": null }`

Response `201 { "conversation_id": "cnv_…", "message": { "role": "assistant", "content": "नमस्ते! आपका व्यवसाय किस तरह का है?", "quick_replies": ["दुकान", "क्लिनिक / ऑफिस", "कई ब्रांच", "फैक्ट्री"] }, "resumed": false }`

Tables: W `conversations`, `messages`, `leads` (an app lead with `origin=app`).

`POST /advisor/message`

Request `{ "conversation_id": "cnv_…", "content": "हमारी 2 ब्रांच हैं, 18 लोग", "modality": "voice" }`

Response `200`:

```
        { "message": { "role": "assistant", "content": "ठीक है। क्या आपके पास GST नंबर है?", "quick_replies": ["हाँ", "नहीं"] },
          "profile": { "segment": "Small multi-site businesses", "employees": 18, "sites": 2, "city_tier": null,
"has_gstin": null, "complete": false, "missing": ["city", "has_gstin", "needs"] },
          "status": "active", "failed_turns": 0, "escalation": null }
```

When the profile completes, `status` becomes `ready_to_recommend`. When `failed_turns` reaches 2 or the user asks for a person, `escalation` = `{ "id": "esc_…", "agent_name": "Priya", "eta": "16:00" }` and `status` = `escalated`.

Errors: `404 CONVERSATION_NOT_FOUND`, `409 CONVERSATION_CLOSED`, `503 PROVIDER_UNAVAILABLE` (a template question is returned instead).

Tables: R/W `conversations`, `messages`; W `escalations` (trigger `advisor_failed` / `user_request`), `notifications`.

`POST /advisor/recommend`

Request `{ "conversation_id": "cnv_…" }`

Response `200`:

```
        { "recommendation_id": "rec_…", "bundle_key": "prof",
          "items": [ { "product_id": "P02", "name": "Biz Fibre 300", "qty": 1, "role": "core", "why": "तेज़ इंटरनेट पूरे ऑफिस
के लिए" } ],
          "addons": [ { "product_id": "P12", "name": "Cloud Telephony / SIP", "qty": 3, "why": "…" } ],
          "blocked": [ { "product_id": "P15", "name": "WhatsApp Business API Starter", "reasons": ["needs an active
GSTIN"], "unlock": "Add your GST number to unlock this." } ],
          "explanation": "2-3 sentences in the user's language", "rules_version": "a1b2c3d" }
```

Errors: `409 PROFILE_INCOMPLETE` (with `missing`), `422 NO_ELIGIBLE_BUNDLE` (with the nearest valid option).

Tables: R `conversations`, `products`; W `recommendations`, `businesses` (profile saved), `messages`.

`GET /advisor/conversations/{id}`

Response `200 { "conversation": {…}, "messages": [ … ], "recommendation": {…} | null }`

Tables: R `conversations`, `messages`, `recommendations`.

`POST /voice/transcribe`

Server fallback. The browser Web Speech API is the default.

Request: multipart `audio` (webm/ogg, ≤ 30 s), `language`.

Response `200 { "text": "…", "confidence": 0.91 }`. Audio is processed in memory and not stored.

Errors: `413 AUDIO_TOO_LONG`, `503 PROVIDER_UNAVAILABLE`. Tables: none.

`POST /voice/speak`

Request `{ "text": "…", "language": "hi" }`

Response `200 audio/mpeg`. Tables: none.

D3.3 Products and quotes

`GET /products`

Query: `?segment=prof&city_tier=2` (optional, filters by catalogue eligibility only).

Response `200 { "products": [{ "product_id": "P01", "name": "Biz Fibre 100", "category": "Connectivity", "unit": "per site", "price_min_inr": 999, "price_max_inr": 1199, "setup_fee_inr": 0, "min_contract_months": 12, "install_days": 5, "requires_gstin": false, "requires_one_of": [], "city_tiers": [1,2], "min_employees": 1, "min_sites": 1 }] }`

Tables: R `products`.

`GET /products/{id}`

Same shape, single item. `404 NOT_FOUND`. Tables: R `products`.

`POST /quotes`

Create from a recommendation.

Request `{ "recommendation_id": "rec_…" }`

Response `201`:

```
        { "quote_id": "quo_…", "status": "draft",
          "items": [ { "product_id": "P02", "name": "Biz Fibre 300", "qty": 1, "role": "core", "selected": true,
"unit_price_inr": 1649, "line_monthly_inr": 1649, "line_setup_inr": 0, "locked": true, "lock_reason": "Needed by Static
IP" } ],
          "monthly_total_inr": 6055, "setup_total_inr": 0, "contract_months": 12, "max_install_days": 5,
          "current_spend_inr": 2600, "valid_until": "2026-07-22" }
```

Errors: `404 RECOMMENDATION_NOT_FOUND`, `409 RECOMMENDATION_STALE` (rules version changed; re-recommend).

Tables: R `recommendations`, `products`, `businesses`; W `quotes`, `quote_items`.

`GET /quotes/{id}`

Same shape as above. Tables: R `quotes`, `quote_items`, `products`.

`PATCH /quotes/{id}`

Sets what the owner pays today, which feeds the “Compared with what you pay now” line. Added in build Phase 2, because the advisor is capped at 6 questions and doesn’t ask about spend.

Request `{ "current_spend_inr": 2500 }`

Response `200`: the full quote, with `current_spend_inr` and `difference_inr` (= monthly total − current spend).

Errors: `400 VALIDATION_ERROR` (negative or above Rs 1 crore), `404 NOT_FOUND`.

Tables: W `quotes`, `businesses` (`monthly_spend_inr`), `leads` (`monthly_telecom_spend_inr`).

Item rules (build Phase 2):

- **What starts selected.** Core items from the recommendation start selected; add-ons start unselected.
- **What locks.** A selected item is locked, with `lock_reason`, if another selected item lists it in `requires_one_of` and no other selected item satisfies that dependency (“Needed by Static IP.”). The only selected internet connection is also locked (“Every plan needs an internet connection.”). Other core items can be switched off.

- **What re-runs.** Every change re-runs `validate_bundle` on the selected set.
- **Steppers.** Shown only for `per_site` items (1 to number of sites) and `per_user` items (1 to employees), and only when that range has more than one value.
- **Order on screen.** Items are listed core first, connection first.

`PATCH /quotes/{id}/items/{product_id}`

Toggle an add-on or change quantity.

Request `{ "selected": false }` or `{ "qty": 4 }`

Response `200`: the full quote (same shape), recalculated.

Errors:

- `409 DEPENDENCY_LOCKED` (`details.reason`: "Static IP needs Biz Fibre")
- `422 INELIGIBLE` (`details.reasons` from the rules engine)
- `409 QUOTE_NOT_DRAFT`
- `422 QTY_NOT_EDITABLE` (fixed-quantity item), `422 QTY_OUT_OF_RANGE` (`details.min`, `details.max`)

Tables: R `products`, `businesses`; W `quote_items`, `quotes`.

`POST /quotes/{id}/accept`

Request `{}`. Response `200 { "quote_id": "quo_…", "status": "accepted", "order_id": "ord_…" }`. This creates the order (same as `POST /orders`).

Errors: `409 QUOTE_EXPIRED`, `422 INELIGIBLE` (re-validated at accept time).

Tables: W `quotes`, `orders`, `order_events`, `notifications` (template `quote_ready`).

D3.4 Orders and tracker

`POST /orders`

Request `{ "quote_id": "quo_…" }`

Response `201 { "order_id": "ord_…", "display_id": "UD-2607-0042", "state": "QUOTE_ACCEPTED", "next_step": "consent" }`

Errors: `409 ORDER_EXISTS` (returns the existing id), `409 QUOTE_NOT_ACCEPTABLE`.

If the bundle is multi-site with SD-WAN, or a manufacturer with a leased line, the order is still created, and an `assisted_deal` escalation assigns an account manager.

Tables: R `quotes`; W `orders`, `order_events`, `escalations` (assisted deals).

`GET /orders` and `GET /orders/{id}`

Response `200 { "orders": [ { "order_id": "…", "display_id": "…", "state": "INSTALL_BOOKED", "monthly_total_inr": 6055, "created_at": "…" } ] }`. The single-order form includes items and `kyc_case_id`.

Tables: R `orders`, `quotes`, `quote_items`, `kyc_cases`.

`GET /orders/{id}/tracker`

Response `200`:

```
        { "order_id": "ord_…", "state": "INSTALL_BOOKED",
          "steps": [
            { "key": "KYC_VERIFIED", "label": "Documents verified", "status": "done", "at": "2026-07-15T11:02:00Z" },
            { "key": "INSTALL_BOOKED", "label": "Installation booked", "status": "current", "at": "2026-07-
15T11:05:00Z", "detail": "Technician visit 18 Jul, 10-12 am" },
            { "key": "INSTALLED", "label": "Installed", "status": "upcoming", "expected": "2026-07-18" },
            { "key": "LIVE", "label": "Live", "status": "upcoming", "expected": "2026-07-19" } ],
          "delayed": false,
          "escalation": { "id": "esc_…", "agent_name": "Priya", "message": "Priya from Accel will call you by 4 pm" }
}
```

Tables: R `orders`, `order_events`, `escalations`, `agents`.

`POST /orders/{id}/pay`

Mock gateway.

Request `{ "method": "upi" }`. Response `200 { "payment_status": "paid" }`.

Errors: `402 PAYMENT_FAILED` (simulated with `?fail=true`, which triggers a retry message).

Tables: W `orders`, `order_events`.

`POST /admin/orders/{id}/advance`

Demo simulator; admin only.

Request `{ "to_state": "INSTALLED", "delay_hours": 0 }`. `delay_hours > 0` back-dates `state_entered_at` so the SLA job fires.

Response `200 { "order_id": "…", "state": "INSTALLED" }`. Errors: `409 INVALID_TRANSITION`.

Tables: W `orders`, `order_events`, `notifications`.

D3.5 Consent

`GET /consent/notice`

Query `?purpose=kyc_verification&language=hi`

Response `200 { "notice_version": "kyc-v1.0", "title": "…", "sections": [{ "heading": "What we collect", "body": "GST certificate, Udyam certificate, signatory ID, address proof" }, { "heading": "Why", "body": "…" }, { "heading": "How long we keep it", "body": "90 days after verification, then deleted" }, { "heading": "Who can see it", "body": "…" }, { "heading": "Your rights", "body": "Withdraw any time…" }] }`

Tables: none (versioned static content).

`POST /consent`

Request `{ "purpose": "kyc_verification", "order_id": "ord_…", "notice_version": "kyc-v1.0", "granted": true }`

Response `201 { "consent_id": "cns_…", "kyc_case_id": "kyc_…", "order_state": "CONSENT_GIVEN" }`

Errors: `409 INVALID_STATE` (order not at QUOTE_ACCEPTED), `400 NOTICE_VERSION_OUTDATED`.

Tables: W `consents`, `kyc_cases`, `orders`, `order_events`.

`DELETE /consent/{id}`

Withdraw consent.

Response `200 { "withdrawn_at": "…", "kyc_case_status": "rejected", "documents_scheduled_for_deletion": 4 }`

Tables: W `consents`, `kyc_cases`, `kyc_documents`, `orders`, `order_events`.

D3.6 KYC

`GET /kyc/gstin/{gstin}`

Response `200 { "gstin_masked": "27•••••••••1Z5", "valid_format": true, "checksum_ok": true, "found": true, "legal_name": "Neha Banerjee (Prop. Neelgiri Physio Clinic)", "trade_name": "Neelgiri Physio Clinic", "constitution": "Proprietorship", "state": "Delhi", "status": "Active", "effects": { "removed_products": [], "quote_repriced": false } }`

If `status` ≠ Active and a quote is in progress, GST-only items are removed, the quote is re-priced, and `effects` lists the changes.

Errors:

- `400 GSTIN_FORMAT` (not 15 characters / wrong pattern)

- `400 GSTIN_CHECKSUM` (check digit wrong; the validator accepts `X` as the 4th PAN character)
- `404 GSTIN_NOT_FOUND`

Tables: R `gstin_registry`; W `businesses` (`gstin_enc`, `gst_status`), `quote_items`/`quotes` (when repriced), `audit_log`.

`GET /kyc/udyam/{num}`

Response `200 { "udyam_masked": "UDYAM-DL-00-•••4567", "found": true, "enterprise_name": "…", "enterprise_type": "Micro", "city": "New Delhi", "state": "Delhi", "matches_gst_trade_name": true, "matches_gst_state": true }`

Errors: `400 UDYAM_FORMAT` (must match `UDYAM-XX-00-NNNNNNN`; district `00` is accepted), `404 UDYAM_NOT_FOUND`.

Tables: R `udyam_registry`, `businesses`; W `businesses` (`udyam_enc`).

`POST /kyc/upload`

Multipart: `case_id`, `doc_type`, `file` (jpg/png/pdf, ≤ 8 MB). OCR runs synchronously, in under about 5 s per page.

Response `201 { "document_id": "doc_…", "doc_type": "address_proof", "version": 2, "quality": { "sharpness": 412.3, "skew_corrected_deg": 0, "ocr_confidence": 92.1, "readable": true }, "extracted": { "bill_date": "2026-06-30", "billed_to": "Trinity Bake House Pvt Ltd" }, "instant_checks": [{ "check_code": "BILL_AGE_90D", "result": "pass" }] }`

Problems found on upload come back immediately as `instant_checks` failures with `message_customer`, so the user can fix them in the same session (e.g. “This photo is blurry…”, “Your bill is from Oct 2025…”).

Errors: `403 CONSENT_REQUIRED`, `409 CASE_NOT_OPEN`, `413 FILE_TOO_LARGE`, `415 UNSUPPORTED_TYPE`.

Tables: R `kyc_cases`, `consents`; W `kyc_documents`, `kyc_checks`.

`POST /kyc/{case_id}/submit`

Request `{}`. Response `202 { "case_id": "kyc_…", "status": "submitted" }`. Checks run in a background task, in under about 10 s.

Errors: `409 DOCUMENTS_MISSING` (`details.missing`: [“udyam_certificate”]). The user can submit without a Udyam certificate only if `has_udyam=false`.

Tables: W `kyc_cases`, `orders` (KYC_SUBMITTED), `order_events`.

`GET /kyc/{case_id}/result`

Response `200`:

```
        { "case_id": "kyc_…", "status": "flagged", "verdict": "Re-upload needed",
          "checks": [ { "check_code": "BILL_AGE_90D", "result": "fail", "severity": "stale", "message": "Your bill is
from 31 Oct 2025. Please upload one from the last 3 months.", "action": { "type": "reupload", "doc_type":
"address_proof" } } ],
          "order_state": "KYC_FLAGGED", "escalation": null }
```

The `verdict` mapping follows the verdicts in [K]:

Verdict	Order state	Extra
Verified	KYC_VERIFIED	
Re-upload needed	KYC_FLAGGED	`action` set per check
Human review	HUMAN_REVIEW	escalation returned
Blocked	KYC_FLAGGED	`action.type = "change_bundle"`

Tables: R `kyc_cases`, `kyc_checks`, `orders`, `escalations`; W `orders`, `order_events`, `escalations`, `notifications`.

D3.7 Escalations

`POST /escalations`

The customer taps “Talk to a person”.

Request `{ "order_id": "ord_…", "conversation_id": null, "reason": "Want to change installation date", "preferred_time": "after 4 pm" }`

Response `201 { "escalation_id": "esc_…", "status": "assigned", "agent": { "name": "Priya", "languages": ["hi","en"] }, "sla_due_at": "…", "message": "Priya from Accel will call you by 4 pm" }`

Errors: `409 ESCALATION_EXISTS` (returns the open one), `429 RATE_LIMITED`.

Tables: R `agents`, `escalations`; W `escalations`, `notifications`, `orders` (`assigned_agent_id`).

`GET /escalations`

Agent or admin.

Query `?status=open,assigned&mine=true&trigger=kyc_flag`

Response `200 { "items": [{ "escalation_id": "…", "trigger": "kyc_flag", "priority": "P1", "status": "assigned", "business": "Saffron Unisex Salon", "reason": "GSTIN on certificate differs from registry", "order_display_id": "UD-2607-0042", "age_minutes": 37, "sla_due_at": "…", "agent": "Priya" }], "counts": { "open": 3, "assigned": 5, "breached": 1 } }`

Tables: R `escalations`, `orders`, `businesses`, `agents`.

`GET /escalations/{id}`

Response `200`: the escalation plus full context (masked profile, chat summary, quote, KYC checks, documents list with view links, order events). No customer has to repeat themselves.

Tables: R `escalations`, `conversations`, `messages`, `quotes`, `quote_items`, `kyc_cases`, `kyc_checks`, `kyc_documents`, `order_events`; W `audit_log`.

`PATCH /escalations/{id}`

Request `{ "action": "assign" | "start" | "resolve" | "reassign", "agent_id": "agt_…", "resolution": "approved", "note": "Confirmed GSTIN with owner; typo on certificate" }`

Response `200`: the escalation. Resolving a `kyc_flag` with `approved` moves the order to KYC_VERIFIED. With `rejected`, the order goes back to KYC_FLAGGED with a message.

Errors: `409 INVALID_STATE`, `403 FORBIDDEN` (not assigned to you).

Tables: W `escalations`, `orders`, `order_events`, `kyc_cases`, `notifications`, `audit_log`.

`GET /kyc/documents/{document_id}/view`

Agent/admin only. Returns a short-lived signed URL.

Response `200 { "url": "/files/…?sig=…", "expires_in": 120 }`

Tables: R `kyc_documents`; W `audit_log`.

D3.8 Admin: leads, metrics, agents

`GET /admin/leads`

Query `?bucket=High&status=open&advisor_outcome=abandoned,escalated&segment=prof&source=Website form&not_contacted=true&sort=-propensity_score&page=1&page_size=50`

Response `200 { "items": [{ "lead_id": "L00421", "business_name": "…", "contact": "Ritu S.", "segment": "Professional offices", "city": "Pune", "stage_reached": "Needs assessed", "drop_reason": null, "advisor_outcome": "abandoned", "propensity": { "score": 0.71, "bucket": "High", "bucket_conversion_pct": 15.9, "reasons": ["+ came via Referral", "+ has a GSTIN", "- tier 2 city"] }, "last_contacted_at": null }], "total": 812 }`

Tables: R `leads`.

`GET /admin/leads/{id}`

Response `200 { "lead": {…masked…}, "propensity": {…}, "conversation_summary": "…" | null, "quote": {…} | null, "calls": [ … ], "notes": [ … ] }`

Tables: R `leads`, `conversations`, `quotes`, `call_logs`, `lead_notes`; W `audit_log`.

`POST /admin/leads/{id}/call`

Request `{ "mode": "twilio" | "mock" }`

Response `201 { "call_id": "call_…", "status": "ringing" | "logged" }`. The outcome is set later with `PATCH /admin/calls/{call_id}` and body `{ "outcome": "interested", "duration_s": 140 }`. "Interested" sends the advisor link by WhatsApp if the lead has opted in.

Errors: `409 RECENTLY_CALLED` (within 24 h), `503 PROVIDER_UNAVAILABLE` (falls back to mock).

Tables: R `leads`; W `call_logs`, `leads` (`last_contacted_at`, `lead_status`), `notifications`.

`PATCH /admin/calls/{call_id}`

Request `{ "outcome": "interested", "duration_s": 140, "note": "Wants Office Pro, will finish KYC tonight" }`

Response `200 { "call_id": "call_…", "outcome": "interested", "lead_status": "contacted", "advisor_link_sent": true }`

Errors: `404 NOT_FOUND`, `400 VALIDATION_ERROR` (unknown outcome).

Tables: W `call_logs`, `leads`, `lead_notes` (if a note is given), `notifications`.

`POST /admin/leads/{id}/notes`

Request `{ "body": "Owner busy until Monday, call after 5 pm" }`. Response `201 { "note_id": "…" }`. Tables: W `lead_notes`.

`GET /admin/metrics`

Query `?from=2026-07-01&to=2026-07-31&segment=all&channel=self_serve|assisted|all`

Response `200`. Baselines come from [B]; `current` is computed from `orders`, `order_events`, `leads`, `call_logs` and `escalations`:

```
{ "metrics": [
  { "key": "lead_to_activation", "baseline": 0.074, "target": 0.11, "current": 0.0 },
  { "key": "days_to_activate", "baseline": 17.6, "target": 8, "current": null },
  { "key": "cac_per_activation", "baseline": 15053, "target": 8000, "current": null },
  { "key": "churn_6m", "baseline": 0.152, "target": 0.11, "current": null },
  { "key": "chase_calls_per_customer", "baseline": 3.0, "target": 0.5, "current": null },
  { "key": "kyc_completion", "baseline": 0.683, "target": 0.85, "current": null },
  { "key": "escalation_resolution_hours", "baseline": null, "target": 24, "current": null } ] }
```

Tables: R `leads`, `orders`, `order_events`, `kyc_cases`, `escalations`, `notifications`.

`GET /admin/agents`

Response `200 { "agents": [ { "id": "agt_…", "name": "Priya", "languages": ["hi","en"], "skills": ["kyc","sales"], "open_cases": 3, "max_open": 8 } ] }`

Tables: R `agents`, `escalations`.

D3.9 Notifications

`POST /notifications/whatsapp`

Internal or admin send, used by services and for demo resends.

Request `{ "user_id": "usr_…", "order_id": "ord_…", "template": "install_booked", "language": "hi", "params": { "date": "18 Jul", "slot": "10-12 am" } }`

Response `202 { "notification_id": "ntf_…", "status": "queued" | "skipped_no_optin" }`

Errors: `404 TEMPLATE_NOT_FOUND`, `422 PARAMS_MISSING`.

Tables: R `users`, `consents`; W `notifications`.

`GET /notifications`

The customer sees their own messages; the admin sees the outbox (`?scope=all`).

Response `200 { "items": [ { "id": "…", "channel": "whatsapp", "template": "kyc_fix_needed", "status": "sent", "preview": "Your bill is older than 3 months…", "created_at": "…" } ] }`

Tables: R `notifications`.

`POST /notifications/webhook/twilio`

Delivery status callback, validated by the Twilio signature.

Request: form-encoded Twilio payload. Response `204`.

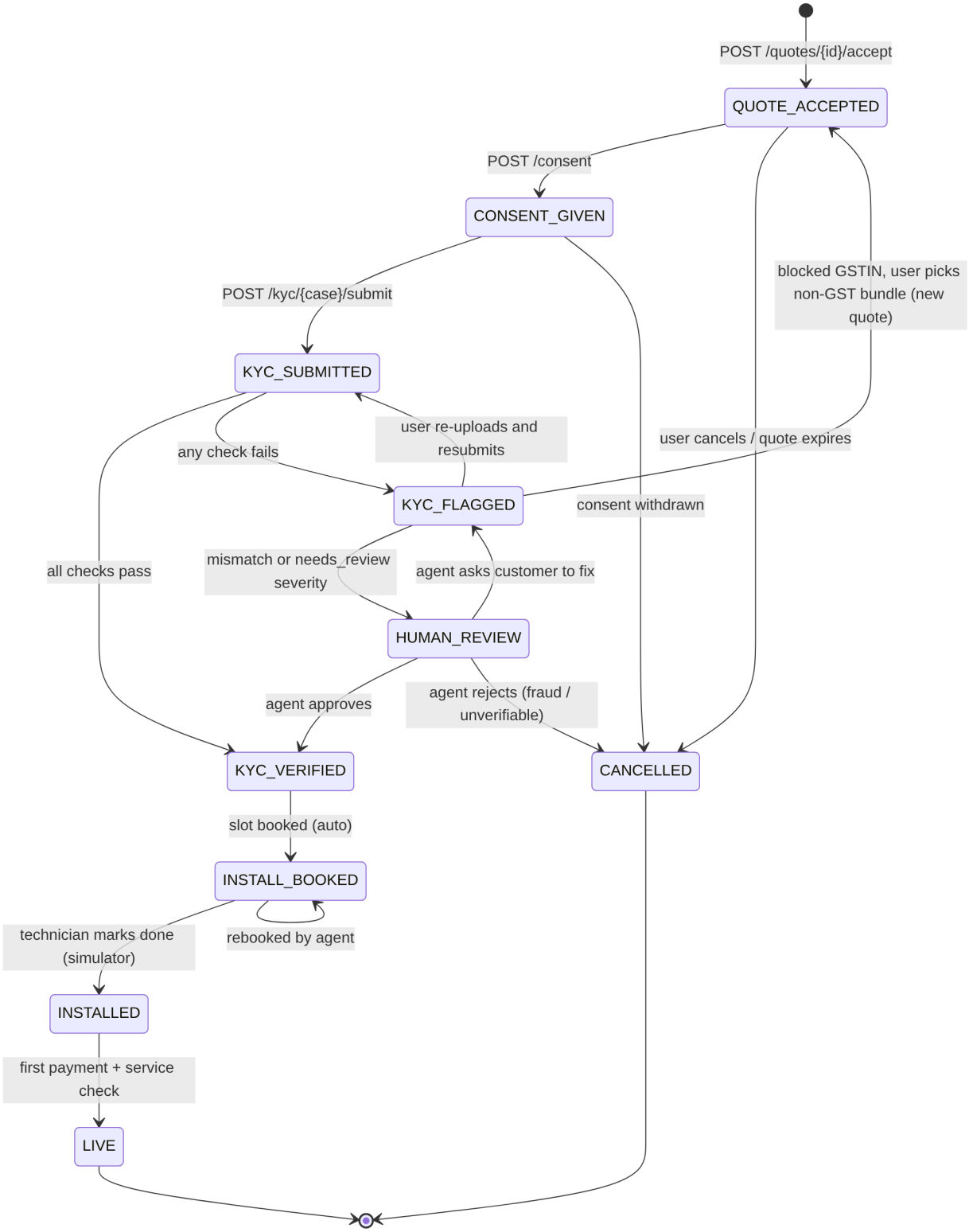
Tables: W `notifications`.

D3.10 System

Endpoint	Purpose	Tables
`GET /health`	Liveness, DB check, provider status (LLM, OCR, messaging: up / degraded / mock)	R `products` (count)
`POST /admin/seed` (dev only)	Loads the 4 CSVs, 5 agents and demo users	W `products`, `leads`, `gstin_registry`, `udyam_registry`, `agents`, `users`

D4. Order state machine and escalation rules

D4.1 States and transitions



Rules:

- **Invalid transitions** return `409 INVALID_TRANSITION`.
- **Every transition writes three things:** an `order_events` row, an updated `orders.state_entered_at`, and a recomputed `state_due_at`. It also queues the matching notification, if the user has opted in.
- **Blocked GSTIN.** The `KYC_FLAGGED → QUOTE_ACCEPTED` path for a blocked GSTIN supersedes the old quote and creates a new one without GST-only products. The KYC documents already uploaded are kept.

D4.2 Expected dates

Date	Formula	Example (Office Pro)
`expected_install_date`	`kyc_verified_at` + max(`install_days` of selected items)	P02 5 d, P08 1 d, P16 1 d, P20 1 d, P17 1 d → +5 days
`expected_live_date`	`expected_install_date` + 1 day	+6 days from verification
Shown before purchase	On the quote, as “Live in about N days after documents are verified”	N = 6

Install days come from `typical_install_days` in the catalogue. Examples:

- Dukaan Digital: 5 days, or 2 with the AirFiber fallback
- Factory Line: 21 days (leased line)
- SD-WAN add-on: 14 days

These are the dates the tracker shows. The target of 8 days from lead to live [B] is only reachable when the install takes 5 days or less. This is one more reason to launch with professional offices.

D4.3 SLA per state (what “late by X hours” means)

State	Expected duration	Late after	Action when late
QUOTE_ACCEPTED (waiting for consent)	minutes	24 h	Reminder (`kyc_reminder_24h`), no escalation. At 7 days, quote expires → CANCELLED
CONSENT_GIVEN (waiting for uploads)	minutes	24 h, then 72 h	Reminder at 24 h. At 72 h, escalation `sla_breach` P3 (agent offers help)
KYC_SUBMITTED	under 10 seconds	1 h	Escalation `sla_breach` P1 (pipeline stuck)
KYC_FLAGGED (waiting for re-upload)	—	48 h	Reminder at 24 h; escalation P3 at 48 h
HUMAN_REVIEW	—	4 business hours to first response	Escalation is created on entry; breach raises priority to P1 and alerts the admin
KYC_VERIFIED (waiting for slot)	instant (auto)	24 h	Escalation P2
INSTALL_BOOKED	until `expected_install_date`	expected_install_date + 24 h	Escalation `sla_breach` P1; customer told “Priya will call you by …”
INSTALLED (waiting for live)	1 day	48 h	Escalation P2

The `SLA_SCAN_INTERVAL_MIN` job runs every 5 minutes. It selects orders where `state_due_at < now` that have no open escalation of the same trigger.

D4.4 Escalation triggers and assignment

Trigger	Fires when	Priority	Required skill
`sla_breach`	Any row in D4.3 crosses its “late after”	per table	install (fulfilment states), kyc (KYC states)
`kyc_flag`	Verdict = Human review (severity `mismatch` / `needs_review`), e.g. K06 GSTIN differs, K07 name names another business [K]	P1	kyc
`advisor_failed`	`conversations.failed_turns` reaches 2	P2	sales
`user_request`	`POST /escalations`, or the user says they want a person in chat	P2	sales (pre-order) / install (post-order)

Trigger	Fires when	Priority	Required skill
`assisted_deal`	Order contains SD-WAN (P10), a leased line (P05/P06), or 3+ sites	P2	account_manager

Assignment algorithm:

1. Filter to `agents.active = 1` with the required skill.
2. Prefer agents whose `languages` include the customer's language.
3. Pick the agent with the fewest open escalations, as long as they're below `max_open`.
4. Break ties by the oldest `assigned_at`.
5. If no agent qualifies, leave the escalation `open` (unassigned) and show it at the top of the admin queue.

On assignment:

- Set `escalations.assigned_agent_id` and `sla_due_at`.
- Set `orders.assigned_agent_id`, if there is an order.
- Send the customer `agent_assigned` (WhatsApp if opted in, plus an in-app banner on the tracker).

D4.5 Notifications per event

Event	Template	Channel
Quote accepted	`quote_ready`	in_app + WhatsApp
Incomplete KYC after 24 h	`kyc_reminder_24h`	WhatsApp (opt-in only)
KYC verified	`kyc_verified`	in_app + WhatsApp
KYC needs a fix	`kyc_fix_needed` (lists the exact fix)	in_app + WhatsApp
KYC in human review	`kyc_review`	in_app + WhatsApp
Install booked	`install_booked` (date, slot)	in_app + WhatsApp
Delay or escalation	`delay_escalated` / `agent_assigned`	in_app + WhatsApp
Live	`live`	in_app + WhatsApp

When the user hasn't opted in, WhatsApp messages are recorded with status `skipped_no_optin`, and the in-app message still shows.

D5. Screen-by-screen UI plan

Layout and theme. Mobile-first at a 360 px width, with a desktop admin console. The theme is Accel Business red and white, with large tap targets of at least 44 px. The real Airtel logo or name is never shown in the product UI.

Components. These names are used in the frontend folder (D6):

- `ChatBubble`
- `QuickReplies`
- `MicButton`
- `BundleCard`
- `PriceBar`
- `AddonToggle`
- `StepTimeline`
- `UploadTile`
- `CheckResult`
- `HumanBanner`
- `DataTable`
- `PropensityBadge`
- `MetricTile`

D5.1 Customer screens

C1. Language pick (`/`)

- **Purpose:** choose the language first, so every following word is understood.
- **Components:** 7 language tiles (हिंदी, English, मराठी, ಕನ್ನಡ, ગુજરાતી, తెలుగు, தமிழ்), auto-highlighting the browser language.
- **API calls:** none. Saved locally, then sent at login. `PATCH /me` if already logged in.
- **Empty/error states:** none. Defaults to Hindi if nothing is chosen.

C2. Login (`/login`)

- **Purpose:** identify the owner with the least friction.
- **Components:** phone input (+91), “Send OTP”, 6-box OTP input, WhatsApp opt-in checkbox (“Send me updates on WhatsApp”), link to the privacy summary.
- **API calls:** `POST /auth/otp/request`, `POST /auth/otp/verify`.
- **Empty/error states:** invalid phone (inline); wrong OTP (shake, attempts left); expired OTP (“Resend”); rate limited (“Try again in 10 minutes”). In mock mode, the dev code is shown in a banner.

C3. Advisor chat (`/advisor`)

- **Purpose:** discover the business in 5–6 questions, by text or voice.
- **Components:**
  - `ChatBubble` list
  - `QuickReplies` chips for common answers
  - `MicButton` (hold to talk, shows the live transcript)
  - progress dots (“question 3 of 6”)
  - “Talk to a person” link in the header
  - read-aloud toggle
- **API calls:**
  - `POST /advisor/conversations`
  - `POST /advisor/message`
  - `POST /voice/transcribe` and `POST /voice/speak`, only when browser speech is unavailable
  - `GET /advisor/conversations/{id}` (resume)
- **Empty/error states:**
  - First visit shows a greeting and the first question.
  - Mic permission denied → “Type instead” hint.
  - LLM unavailable → the template question flow continues, invisible to the user.
  - Two failed turns → `HumanBanner` “Priya will call you by 4 pm”, and the chat stays open.
  - Network loss → message queued with a retry chip.

C4. Recommendation card (`/advisor/recommendation`)

- **Purpose:** show the bundle and why it fits, in plain words, with no product codes.
- **Components:**
  - `BundleCard` (each item with an icon, name, one-line "why", quantity)
  - "Not available yet" section for blocked items, with the unlock reason (e.g. "Add your GST number to unlock WhatsApp Business")
  - explanation paragraph
  - "See price" button
- **API calls:** `POST /advisor/recommend`, `POST /quotes`.
- **Empty/error states:**
  - `PROFILE_INCOMPLETE` → back to chat with the missing question.
  - `NO_ELIGIBLE_BUNDLE` → nearest valid option plus a "Talk to a person" offer.

C5. Quote with add-on toggles (`/quote/[id]`)

- **Purpose:** one clear monthly price the owner can shape. This targets the 43.6% quote step [F].
- **Components:**
  - sticky `PriceBar` (monthly total, setup fee, contract months)
  - core items (locked, with a lock reason on tap)
  - `AddonToggle` list with price deltas
  - quantity steppers for per-site and per-user items
  - "Compared with what you pay now" line. If the spend isn't known, the owner picks a chip (Rs 500 / 1,000 / 2,500 / 5,000 / 10,000) or types an amount, sent with `PATCH /quotes/{id}`
  - "Live in about N days" line
  - the total updates on screen the moment a toggle or stepper is tapped (computed in the browser from the unit prices), then is confirmed by the server; if the server rejects the change, it rolls back and shows the reason
  - "Accept" button
- **API calls:** `GET /quotes/{id}`, `PATCH /quotes/{id}/items/{pid}`, `PATCH /quotes/{id}`, `POST /quotes/{id}/accept`.
- **Empty/error states:**
  - `DEPENDENCY_LOCKED` → toast with the reason.
  - `INELIGIBLE` → toggle reverts with the reason.
  - `QUOTE_EXPIRED` → "Prices refreshed" and reload.
  - Assisted deal → banner "An account manager will confirm your site survey".

C6. Consent (`/order/[id]/consent`)

- **Purpose:** informed consent before any document is collected (DPDP).
- **Components:** notice sections (what, why, how long, who sees it, your rights), "I agree" checkbox, "Continue" button, "Not now" button.
- **API calls:** `GET /consent/notice`, `POST /consent`.
- **Empty/error states:**
  - "Not now" → saved; reminder after 24 h if opted in.
  - `NOTICE_VERSION_OUTDATED` → reload the notice.

C7. KYC (`/order/[id]/kyc`)

A three-step stepper.

C7a. GSTIN - **Components:** GSTIN input with live format and checksum validation; "I don't have GST" link; registry result card (masked GSTIN, trade name, status). Udyam number input, optional. - **API calls:** `GET /kyc/gstin/{gstin}`, `GET /kyc/udyam/{num}`. - **Errors:** - Format or checksum error inline. - Not found → "Check the number or upload your certificate". - Cancelled or suspended → explainer plus the re-priced bundle without GST-only items.

C7b. Uploads - **Components:** 4 `UploadTile`s (camera or file) with instant quality feedback. The Udyam tile is hidden when `has_udyam=false`. - **API calls:** `POST /kyc/upload` per document. - **Errors:** - Blurry → "Retake in good light". - Stale bill → "Upload a bill from the last 3 months". - Too large or wrong type → inline message. - Consent missing → back to C6.

C7c. Results - **Components:** `CheckResult` list (green ticks, fix-it items with a re-upload button), overall verdict. - **API calls:** `POST /kyc/{case}/submit`, then `GET /kyc/{case}/result` (polled every 2 s until done). - **Empty/error states:** - Submitting → "Checking your documents…" (under 10 s). - Human review → `HumanBanner` "A specialist will check this with you by …". - Blocked → "Choose a plan that doesn't need GST" → new quote.

C8. Order placed (`/order/[id]/placed`)

- **Purpose:** confirm the order and take the first payment (mock).
- **Components:** order number, bundle summary, expected install and live dates, “Pay Rs X” (UPI/card mock), “Track my order” button.
- **API calls:** `GET /orders/{id}`, `POST /orders/{id}/pay`.
- **Empty/error states:** `PAYMENT_FAILED` → retry with an alternative method; leaving the screen keeps the payment pending, with a tracker reminder.

C9. Tracker (`/order/[id]`)

This is the home screen after purchase.

- **Purpose:** remove the need to chase. Churn runs 6% at 1 chase call vs 28% at 5+ [S].
- **Components:**
  - `StepTimeline` (Verified → Installation booked → Installed → Live) with done/current/upcoming states, actual and expected dates
  - plain-language detail (“Technician visit 18 Jul, 10–12 am”)
  - delay banner
  - `HumanBanner` with the agent’s name when escalated
  - “Talk to a person” button
  - WhatsApp updates toggle
- **API calls:** `GET /orders/{id}/tracker` (on open, then every 60 s), `GET /orders`, `GET /notifications`.
- **Empty/error states:**
  - No orders → “Start with the advisor”.
  - Offline → last known status with a timestamp.
  - Delayed → amber banner plus the assigned agent’s name.

C10. Talk to a person (bottom sheet, available from C3, C5, C7, C9)

- **Purpose:** an explicit human path at any point.
- **Components:** reason picker (pre-filled from context), preferred time, language, “Request call” button; then a confirmation card with the agent’s name and ETA.
- **API calls:** `POST /escalations`.
- **Empty/error states:**
  - `ESCALATION_EXISTS` → show the existing one.
  - No agent free → “We’ll call you by tomorrow 11 am” (open queue).

D5.2 Admin and agent screens (`/admin`, desktop-first)

A1. Admin login

- **Purpose:** staff access.
- **Components:** the same OTP form; role checked.
- **API calls:** `POST /auth/otp/request`, `POST /auth/otp/verify`, `GET /me`.
- **Errors:** `FORBIDDEN` for customer accounts.

A2. Leads table (`/admin/leads`)

- **Purpose:** decide whom to call first. The top 20% of leads hold 43% of activations [M].
- **Components:**
  - filter row: bucket, segment, source, advisor outcome, “not contacted”, date range
  - `DataTable` sorted by propensity, with a `PropensityBadge` (bucket + “converts ~15.9%”) and top 3 reasons on hover
  - masked contact
  - metric strip of 5 `MetricTile`s (baseline → target → current)
- **API calls:** `GET /admin/leads`, `GET /admin/metrics`.
- **Empty/error states:** no matches → “No leads match these filters” with a reset button; metrics with no app data yet → “—” with “baseline from Apr–Jun 2026”.

A3. Lead detail (`/admin/leads/[id]`)

- **Purpose:** call with context and record the outcome.
- **Components:** profile card (masked), propensity with reasons, chat summary and abandoned quote (if any), “Call” button, outcome picker, notes list with an add-note box, call history.
- **API calls:** `GET /admin/leads/{id}`, `POST /admin/leads/{id}/call`, `PATCH /admin/calls/{call_id}`, `POST /admin/leads/{id}/notes`.
- **Empty/error states:**
  - `RECENTLY_CALLED` → “Called 3 h ago by Priya”.
  - Call provider down → mock call logged with a notice.
  - No conversation → “Imported lead, no app activity”.

A4. Escalations queue (`/admin/escalations`)

- **Purpose:** humans handle exceptions.
- **Components:** tabs (Mine, Open, All), list with trigger icon, priority, SLA countdown (red when breached), business, reason, agent; “Assign to me” and reassign actions.
- **API calls:** `GET /escalations`, `PATCH /escalations/{id}` (assign/reassign), `GET /admin/agents`.
- **Empty/error states:** empty → “No open escalations” (a good state); breached items are pinned to the top.

A5. Escalation / KYC review detail (`/admin/escalations/[id]`)

- **Purpose:** resolve a case without making the customer repeat anything.
- **Components:**
  - context panel: profile, chat summary, quote, order timeline
  - KYC checks with severities
  - document viewer (signed URL, audit-logged)
  - side-by-side “certificate says / registry says” for mismatches (K06, K07)
  - resolution buttons (Approve, Ask customer to fix, Reject, Rebook install) with a required note
- **API calls:** `GET /escalations/{id}`, `GET /kyc/documents/{id}/view`, `PATCH /escalations/{id}`.
- **Empty/error states:** document deleted after retention → “Document deleted on [date] (retention)”; `INVALID_STATE` → refresh.

A6. Notifications outbox (`/admin/notifications`)

- **Purpose:** show in the demo that messages went out, even in mock mode.
- **Components:** table of channel, template, status, preview, time; “Resend” action.
- **API calls:** `GET /notifications?scope=all`, `POST /notifications/whatsapp`.
- **Empty/error states:** empty → “No messages yet”; failed rows in red with the error text.

A7. Order simulator (`/admin/simulator`)

Demo only.

- **Purpose:** move orders through states and create delays on cue for the demo video.
- **Components:** order picker, “Advance to…” buttons, “Make it late by N hours” input, live event log.
- **API calls:** `GET /orders` (admin scope), `POST /admin/orders/{id}/advance`, `GET /orders/{id}/tracker`.
- **Empty/error states:** `INVALID_TRANSITION` → shows the allowed next states.

D6. Folder structure and environment

D6.1 Repository layout

```
udyam-digital/
├── data/                        # supplied CSVs + kyc_documents/ (git-ignored in public repo)
├── analysis/                    # Phase 1 scripts and outputs (already done)
├── docs/                        # PHASE_1 report, DESIGN.md, figures/, privacy note, deck
├── backend/
│   ├── pyproject.toml
│   ├── alembic/                 # migrations (or create_all for the demo)
│   ├── models/propensity.pkl
│   ├── app/
│   │   ├── main.py              # FastAPI app, routers, middleware
│   │   ├── config.py            # pydantic-settings, reads env
│   │   ├── db.py                # engine, session
│   │   ├── security/            # jwt.py, crypto.py (Fernet, HMAC), masking.py, log_filter.py
│   │   ├── models/              # SQLAlchemy: user.py, business.py, lead.py, product.py, registry.py,
│   │   │                        #   conversation.py, recommendation.py, quote.py, order.py,
│   │   │                        #   consent.py, kyc.py, escalation.py, notification.py, call.py, audit.py
│   │   ├── schemas/             # Pydantic request/response models per router
│   │   ├── routers/             # auth.py, advisor.py, voice.py, products.py, quotes.py, orders.py,
│   │   │                        #   consent.py, kyc.py, escalations.py, admin.py, notifications.py, health.py
│   │   ├── rules/               # engine.py (from validate_bundles.py), bundles.json
│   │   ├── pricing/service.py
│   │   ├── advisor/             # orchestrator.py, prompts/, llm/ (base.py, provider adapters), guards.py
│   │   ├── kyc/                 # ocr.py, gstin.py (checksum + repair), checks.py, names.py, storage.py
│   │   ├── orders/              # state_machine.py, tracker.py, dates.py
│   │   ├── escalations/         # triggers.py, assign.py, sla_job.py
│   │   ├── notifications/       # service.py, templates/{hi,en,mr,kn,gu,te,ta}.json, providers/ (twilio.py, mock.py)
│   │   ├── voice/               # base.py, providers/
│   │   ├── propensity.py
│   │   └── seed.py              # loads CSVs, agents, demo users
│   └── tests/
│       ├── test_rules.py        # 22 bundle checks + 9 must-refuse (from Phase 1)
│       ├── test_pricing.py
│       ├── test_gstin.py        # checksum, 'X' 4th PAN char, repair
│       ├── test_kyc_applicants.py# K01–K12 expected verdicts
│       ├── test_state_machine.py
│       ├── test_escalations.py
│       ├── test_advisor_scenarios.py  # our 15 judge-style scenarios
│       └── test_privacy.py      # no raw IDs in logs/responses
└── frontend/
    ├── package.json
    ├── next.config.js
    ├── tailwind.config.ts       # Accel red/white tokens
    ├── messages/                # hi.json, en.json, mr.json, kn.json, gu.json, te.json, ta.json
    ├── public/                  # icons, illustrations (no third-party logos)
    ├── src/
    │   ├── app/
    │   │   ├── page.tsx                      # C1 language
    │   │   ├── login/page.tsx                # C2
    │   │   ├── advisor/page.tsx              # C3
    │   │   ├── advisor/recommendation/page.tsx # C4
    │   │   ├── quote/[id]/page.tsx           # C5
    │   │   ├── order/[id]/consent/page.tsx   # C6
    │   │   ├── order/[id]/kyc/page.tsx       # C7a-c
    │   │   ├── order/[id]/placed/page.tsx    # C8
    │   │   ├── order/[id]/page.tsx           # C9 tracker
    │   │   └── admin/
    │   │       ├── login/page.tsx            # A1
    │   │       ├── leads/page.tsx            # A2
    │   │       ├── leads/[id]/page.tsx       # A3
    │   │       ├── escalations/page.tsx      # A4
    │   │       ├── escalations/[id]/page.tsx # A5
    │   │       ├── notifications/page.tsx    # A6
    │   │       └── simulator/page.tsx        # A7
    │   ├── components/          # ChatBubble, QuickReplies, MicButton, BundleCard, PriceBar, AddonToggle,
    │   │                        #   StepTimeline, UploadTile, CheckResult, HumanBanner, TalkToPersonSheet (C10),
    │   │                        #   DataTable, PropensityBadge, MetricTile
    │   ├── lib/                 # api.ts (typed client), auth.ts, speech.ts, i18n.ts, format.ts (Rs, dates)
    │   └── types/               # generated from FastAPI OpenAPI schema
    └── tests/e2e/               # Playwright happy path on a 360 px viewport
```

D6.2 Environment variables

Variable	Used by	Example / note
APP_ENV	backend	dev / demo / prod
DATABASE_URL	backend	sqlite:///./udyam.db
JWT_SECRET, JWT_EXPIRES_MIN	backend	random 32+ bytes; 1440
APP_ENCRYPTION_KEY	backend	Fernet key for PII columns and document keys
PII_HASH_KEY	backend	HMAC key for phone_hash / client_hash
OTP_MODE	backend	mock (returns dev_code) / sms
CORS_ORIGINS	backend	http://localhost:3000,https://<frontend>.vercel.app
LLM_PROVIDER, LLM_MODEL, LLM_API_KEY	advisor	provider adapter; LLM_PROVIDER=template runs without an LLM
LLM_TIMEOUT_S, ADVISOR_MAX_QUESTIONS, ADVISOR_MAX_FAILED_TURNS	advisor	8, 6, 2
OCR_ENGINE, TESSERACT_CMD	KYC	tesseract / paddle; binary path
KYC_TODAY	KYC	2026-07-15 (fixed “today” for the dataset’s date checks)
KYC_ADDRESS_PROOF_MAX_DAYS, KYC_BLUR_THRESHOLD	KYC	90, 50
UPLOAD_DIR, UPLOAD_MAX_MB, DOC_RETENTION_DAYS, DOC_RETENTION_ABANDONED_DAYS	KYC	./storage, 8, 90, 30
NOTIFY_PROVIDER	notifications	mock / twilio
TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN, TWILIO_WHATSAPP_FROM, TWILIO_VOICE_FROM	notifications, calls	sandbox credentials
VOICE_PROVIDER, VOICE_API_KEY	voice	browser (default) / bhashini / sarvam
SLA_SCAN_INTERVAL_MIN, BUSINESS_HOURS	escalations	5, 09:00-19:00 Asia/Kolkata
PROPENSITY_MODEL_PATH	admin	backend/models/propensity.pkl
NEXT_PUBLIC_API_BASE_URL	frontend	http://localhost:8000/api/v1
NEXT_PUBLIC_DEFAULT_LANG, NEXT_PUBLIC_SUPPORTED_LANGS	frontend	hi; hi,en,mr,kn,gu,te,ta
NEXT_PUBLIC_DEMO_MODE	frontend	true shows the mock OTP code and the simulator link

D7. Traceability and coverage checks

D7.1 Screen → API endpoints

Screen	Endpoints
C1 Language pick	`PATCH /me` (if logged in)
C2 Login	`POST /auth/otp/request`, `POST /auth/otp/verify`
C3 Advisor chat	`POST /advisor/conversations`, `POST /advisor/message`, `GET /advisor/conversations/{id}`, `POST /voice/transcribe`, `POST /voice/speak`
C4 Recommendation	`POST /advisor/recommend`, `POST /quotes`
C5 Quote	`GET /quotes/{id}`, `PATCH /quotes/{id}`, `PATCH /quotes/{id}/items/{product_id}`, `POST /quotes/{id}/accept`
C6 Consent	`GET /consent/notice`, `POST /consent`
C7a GSTIN	`GET /kyc/gstin/{gstin}`, `GET /kyc/udyam/{num}`
C7b Uploads	`POST /kyc/upload`
C7c Results	`POST /kyc/{case_id}/submit`, `GET /kyc/{case_id}/result`
C8 Order placed	`GET /orders/{id}`, `POST /orders/{id}/pay`
C9 Tracker	`GET /orders/{id}/tracker`, `GET /orders`, `GET /notifications`, `PATCH /me`
C10 Talk to a person	`POST /escalations`
A1 Admin login	`POST /auth/otp/request`, `POST /auth/otp/verify`, `GET /me`
A2 Leads table	`GET /admin/leads`, `GET /admin/metrics`
A3 Lead detail	`GET /admin/leads/{id}`, `POST /admin/leads/{id}/call`, `PATCH /admin/calls/{call_id}`, `POST /admin/leads/{id}/notes`
A4 Escalations queue	`GET /escalations`, `PATCH /escalations/{id}`, `GET /admin/agents`
A5 Escalation detail	`GET /escalations/{id}`, `GET /kyc/documents/{document_id}/view`, `PATCH /escalations/{id}`
A6 Notifications outbox	`GET /notifications`, `POST /notifications/whatsapp`
A7 Simulator	`GET /orders`, `POST /admin/orders/{id}/advance`, `GET /orders/{id}/tracker`

Endpoints with no screen of their own are backend or integration endpoints, by design:

- `POST /orders` (called by accept)
- `GET /products`, `GET /products/{id}` (advisor and quote internals, public catalogue)
- `DELETE /consent/{id}` (privacy page link)
- `POST /auth/logout`
- `POST /notifications/webhook/twilio`
- `GET /health`
- `POST /admin/seed`

D7.2 API endpoint → tables

Every endpoint in D3 lists its tables. Taking them together, every table in D2 is used by at least one endpoint:

- `users`, `otp_requests`, `agents`, `audit_log`
- `businesses`, `leads`, `products`, `gstin_registry`, `udyam_registry`
- `conversations`, `messages`, `recommendations`
- `quotes`, `quote_items`, `orders`, `order_events`
- `consents`, `kyc_cases`, `kyc_documents`, `kyc_checks`
- `escalations`, `notifications`, `call_logs`, `lead_notes`

The check in D7.4 confirms this mechanically.

D7.3 Coverage of build phases 1–5

Phase	Requirement (from the build prompts)	Where in this design
1 Core	Scaffold + seed script loading 4 CSVs	D6.1 `seed.py`, `POST /admin/seed`
1	Accel theme, 360 px, language picker, i18n	D5 header, C1, D6.1 `messages/`, env `NEXT_PUBLIC_SUPPORTED_LANGS`
1	Phone + OTP (mock), JWT	D3.1, `users`, `otp_requests`, env `OTP_MODE`, `JWT_*`
1	Rules engine with reasons + pytest	D1.1, D1.4, `rules/engine.py`, `test_rules.py`
1	Advisor ≤ 6 questions, explains only eligible items	D1.3 flow 1, D1.4, `POST /advisor/message`, `/advisor/recommend`
1	Recommendation card	C4
2 Conversion	Pricing rule inside min–max, setup, contract	`pricing/service.py`, `quotes`, `quote_items.unit_price_inr`
2	Add-on toggles, dependency lock, quantity steppers	`PATCH /quotes/{id}/items/{pid}`, `quote_items.locked`, C5
2	“Compared with what you pay now”	`quotes.current_spend_inr`, C5
2	Accept → order + event	`POST /quotes/{id}/accept`, `orders`, `order_events`
3 Smart KYC	Consent first, versioned, withdrawable	C6, D3.5, `consents`, `kyc_cases.consent_id NOT NULL`, D2.8
3	GSTIN validator (checksum, ‘X’ PAN char) + mock API	`GET /kyc/gstin/{gstin}`, `kyc/gstin.py`, `test_gstin.py`
3	Udyam validator (district 00) + mock API	`GET /kyc/udyam/{num}`, `udyam_registry` index on name + city
3	OCR of 4 documents	`POST /kyc/upload`, `kyc/ocr.py`, `kyc_documents`
3	Cross-checks with reasons	`kyc_checks.check_code` list, `GET /kyc/{case}/result`
3	Run on K01–K12, results file	`test_kyc_applicants.py`, `analysis/kyc/kyc_results.md` [K]
3	Result screen, re-upload, flag → escalation	C7c, D4.1 KYC_FLAGGED / HUMAN_REVIEW, trigger `kyc_flag`
3	Mask identifiers in UI and logs	D2 conventions, D2.8, `security/masking.py`, `log_filter.py`
4 Operations	Tracker with expected dates	C9, `GET /orders/{id}/tracker`, D4.2
4	Simulator for demo	A7, `POST /admin/orders/{id}/advance`
4	Escalation job, least-loaded + language-matched agent	D4.3, D4.4, `escalations/sla_job.py`, `assign.py`
4	5 seeded agents, agent view with full context	`seed.py`, `agents`, A4, A5, `GET /escalations/{id}`
5 Differentiators	Voice in the advisor	C3 `MicButton`, `POST /voice/*`, `voice/`
5	WhatsApp opt-in + event messages + 24 h reminder	C2 checkbox, D4.5, `notifications`, `POST /notifications/whatsapp`, A6
5	Admin leads with propensity + filters + Call	A2, A3, `GET /admin/leads`, `POST /admin/leads/{id}/call`, `call_logs`
5	Metrics strip with the pilot metrics	A2 `MetricTile`, `GET /admin/metrics`

D7.4 Automated consistency check

`scripts/check_design.py` parses this document and verifies three things:

1. Every endpoint named in D7.1 appears as a D3 heading or in D3.10.

2. Every table in D2 is written or read by at least one D3 endpoint.
3. Every D3 endpoint either appears in D7.1 or is in the listed backend-only set.

The result at the time of writing is in the line below, generated by the script.

```
$ python scripts/check_design.py docs/DESIGN.md
Tables in D2: 24 -> users, otp_requests, agents, audit_log, businesses, leads, products, gstin_registry,
udyam_registry, conversations, messages, recommendations, quotes, quote_items, orders, order_events, consents,
kyc_cases, kyc_documents, kyc_checks, escalations, notifications, call_logs, lead_notes
Endpoints in D3: 49; mapped to screens: 41; backend-only: 8
Tables used by endpoints: 24/24
RESULT: PASS
```